

This material is neither intended to be distributed to Mainland China investors nor to provide securities investment consultancy services within the territory of Mainland China. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan.

Yinlun - A

Capturing growth across AIDC power, liquid cooling and truck electrification; initiate with Overweight

We initiate coverage of Yinlun with an Overweight rating and a Dec-27 PT of Rmb51 (49% upside from last close). We expect the company to deliver ~20% revenue CAGR and ~30% net profit CAGR over 2025-28E. We forecast net profit to more than double by 2028E, while ROE improves from 14% in 2025 to c.20% in 2028E and free cash flow turns materially positive as new businesses scale and capex intensity moderates. We believe the market still largely values Yinlun as a traditional thermal-management supplier, while underappreciating its increasing exposure to AI data center power infrastructure and liquid cooling, two end markets with structurally higher growth and margin profiles than its legacy automotive parts business.

- **AIDC-driven power demand: Genset cooling packages are becoming the key earnings driver.** Yinlun is benefiting from accelerating AI-driven power demand through Caterpillar's (CAT US, OW, covered by Tami Zakaria) backup- and prime-power engine business. CAT highlighted strong visibility extending towards 2030, while J.P. Morgan forecasts CAT's diesel & gas engine capacity growing at ~25% CAGR between 2025 and 2030. At the same time, competitor Modine (MOD US, Not Covered) is increasingly allocating capital and management resources toward data center cooling, creating an opportunity for Yinlun to gain share at CAT through its cost competitiveness, Mexico footprint and capacity expansion. We forecast Yinlun's genset revenue to rise from Rmb0.5bn in 2026E to Rmb3.2bn by 2028E, contributing >10% of Yinlun's revenue and ~20% of group profit.
- **Liquid cooling provides a second AI infrastructure growth leg, with Google-related orders as upside.** Yinlun is the only domestic plate heat exchanger (PHE) supplier among peers with in-house furnace design capability and large-scale manufacturing experience, enabling faster product iteration, capacity expansion and customer qualification. The company passed Google's designated third-party certification in March 2026 and is progressing through CDU integrator validation. Importantly, our base case forecasts do not incorporate potential Google-related orders. Successful qualification and order conversion could provide 4-6% upside to our 2027-28E earnings estimates, with the liquid cooling profit contribution potentially approaching c.10% of group earnings by 2028E.
- **Commercial vehicle (CV) electrification and exports provide a resilient growth foundation.** CV & non-road remain Yinlun's highest-margin legacy segment, supported by export demand, replacement cycles and rising electrification trends. Thermal-management content increases from approximately Rmb14-15k per ICE truck to Rmb20-30k per electric truck, creating roughly a 1.5-2.0x content uplift. We expect HDT EV penetration to increase from ~25% in 2025 to ~50% by 2030E, supporting both revenue growth and content expansion.
- **Initiate at OW.** Our Dec-27 PT of Rmb51 is based on an SOTP and implies 29x/20x blended 2027E/28E P/Es, vs. the current price trading at 18x 2027E.

Initiation

Overweight

002126.SZ, 002126.CH

Price (29 Sep 26): Rmb34.20

Price Target (Dec-27): Rmb51.00

Asia Autos & EV Battery

Rebecca Wen ^{AC}

(852) 2800-8505

rebecca.y.wen@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Shirley Feng

(86-21) 6106-6149

shirley.x.feng@jpmorgan.com

SAC Registration Number: S1730524040001

J.P. Morgan Securities (China) Company Limited

Nick Lai

(65) 6801-3176

nick.yc.lai@jpmorgan.com

J.P. Morgan Securities Singapore Private Limited/

J.P. Morgan Securities (Asia Pacific) Limited/ J.P.

Morgan Broking (Hong Kong) Limited

Style Exposure

Quant Factors	Current	Hist %Rank (1=Top)			
	%Rank	6M	1Y	3Y	5Y
Value	75	84	74	65	54
Growth	14	21	7	5	71
Momentum	45	20	29	10	78
Quality	19	26	15	7	61
Low Vol	86	83	73	84	55

Glossary

AIDC = Artificial Intelligence Data Center

BPHE = Brazen Plate Heat Exchanger

BTMS = Battery Thermal Management System

CDU = Coolant Distribution Unit

CV = Commercial Vehicle

EGR = Exhaust Gas Recirculation

ESS = Energy Storage System

EV = Electric Vehicle

HDT = Heavy-Duty Truck

HVAC = Heating, Ventilation and Air Conditioning

ICE = Internal Combustion Engine

LNG = Liquefied Natural Gas

NEV = New Energy Vehicle

PHE = Plate Heat Exchanger

PV = Passenger Vehicle

Sources for: Style Exposure – J.P. Morgan Global Markets Strategy; all other tables are company data and J.P. Morgan estimates.

See page 43 for analyst certification and important disclosures, including non-US analyst disclosures.

Price Performance



Company Data

Shares O/S (mn)	862
52-week range (Rmb)	62.00-31.70
Market cap (\$ mn)	4,394
Exchange rate	6.71
Free float (%)	83.3%
3M ADV (mn)	20.66
3M ADV (\$ mn)	137.1
Volatility (90 Day)	63
Index	-
BBG ANR (Buy Hold Sell)	19 0 0

Key Metrics (FYE Dec)

Rmb in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	15,678	18,921	22,768	26,633
Adj. EBITDA	2,020	2,196	2,836	3,718
Adj. EBIT	1,269	1,434	2,030	2,887
Adj. net income	957	1,075	1,525	2,173
Net margin	6.1%	5.7%	6.7%	8.2%
Adj. EPS	1.11	1.25	1.77	2.52
BBG EPS	1.17	1.39	1.93	2.55
Cashflow from operations	1,405	1,649	2,492	3,405
FCFF	(102)	435	1,364	2,377
Margins and Growth				
Revenue Growth Y/Y (%)	23.4%	20.7%	20.3%	17.0%
EBITDA margin	12.9%	11.6%	12.5%	14.0%
EBITDA Growth Y/Y (%)	22.4%	8.7%	29.2%	31.1%
EBIT margin	8.1%	7.6%	8.9%	10.8%
Adj. EPS growth	20.7%	12.3%	41.8%	42.5%
Ratios				
Adj. tax rate	13.4%	13.4%	13.4%	13.4%
Interest cover	-	-	-	-
Net debt/Equity	0.6	0.5	0.4	0.2
Net debt/EBITDA	2.3	2.2	1.4	0.7
ROCE	8.2%	8.1%	10.4%	13.1%
ROE	14.1%	13.7%	16.9%	20.6%
Valuation				
FCFF yield	(0.3%)	1.5%	4.6%	8.1%
Dividend yield	0.4%	0.4%	0.6%	1.5%
EV/EBITDA	18.1	16.9	12.9	9.5
EV/Revenue	2.3	2.0	1.6	1.3
Adj. P/E	30.8	27.4	19.3	13.6

Summary Investment Thesis and Valuation

Investment Thesis

Yinlun is a China-based thermal-management supplier with more than four decades of experience in heat exchanger design, brazing and large-scale manufacturing. Its established passenger vehicle and commercial vehicle businesses provide a broad revenue base, while its global manufacturing footprint and long-standing relationships with domestic and international OEMs support expansion into adjacent industrial applications. The key upside comes from AI infrastructure. Yinlun is extending its thermal-management capabilities into data center power generation and liquid cooling, two markets with structurally faster growth and stronger margin potential than its traditional auto parts business. Caterpillar genset cooling packages provide the clearest near-term earnings catalyst, supported by rising AI- and data center-driven power demand and a multi-year supplier qualification barrier. Liquid-cooling PHEs represent a second growth opportunity as higher AI server power density drives CDU demand; Google-related qualification remains potential upside rather than a base case assumption. Together, these businesses should support a material improvement in Yinlun's earnings mix and valuation profile.

Valuation

We have a Dec-27 price target of Rmb51. We use an SOTP valuation, applying a 15x 2028E P/E to the PV and CV auto parts segments and a 35x 2028E P/E to the Digital & Energy segment, reflecting premium growth-stage multiples for AIDC power infrastructure and liquid cooling exposure. This implies a blended 29x 2027E P/E and 20x 2028E P/E.

Performance Drivers

Market	10%
Region	0%
Macro	13%
Style	24%
Idiosyn.	53%

Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.28	0.31
Region: China	-0.14	-0.02
Macro:		
Generic 1st 'CO' Future	-0.38	-0.32
Citi Economic Surprise - EM	-0.21	-0.26
Markit EM Composite PMI SA	0.25	0.21
Quant Styles:		
Value	-0.40	-0.49
DivYld	-0.34	-0.44
Growth	0.36	0.44

Table Of Contents

Executive Summary	4
Investment Thesis	6
1) AIDC-driven power demand: Caterpillar orders support growth visibility	6
2) Liquid cooling creates a second AI data center growth opportunity; Google orders are upside optionality	8
3) Commercial vehicle electrification and exports provide a resilient growth foundation	11
Investment Risks	15
1) High customer concentration in genset business creates significant dependence on Caterpillar	15
2) Liquid-cooling PHE orders have yet to be secured, while capacity is being added ahead of confirmed demand	15
3) Geopolitical risk**this section under GR review**	16
Share Price Analysis.....	17
Valuation	19
Industry	22
Genset: cooling packages for gas/diesel engines; AIDC-driven power infrastructure demand	22
AI server liquid cooling: plate heat exchangers (PHE).....	23
Commercial vehicles: exports and electrification trends to drive demand and content value	25
Company Overview.....	28
Yinlun's 2030 targets.....	35
Financial Analysis	36
Profit and loss analysis	36
Balance sheet and cash flow analysis.....	39

Executive Summary

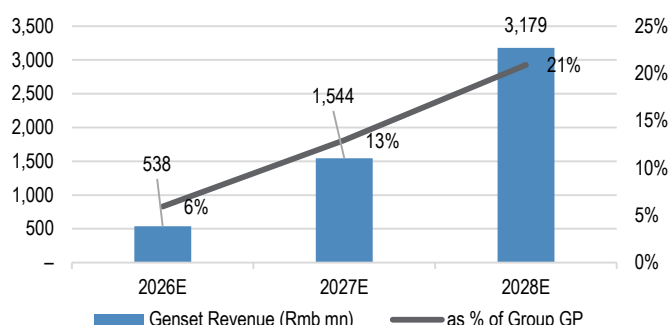
We initiate coverage of Yinlun with an Overweight rating and a Dec-27 price target of Rmb51 (49% upside from last close). Our Dec-27 PT of Rmb51 is based on an SOTP and implies 29x/20x blended 2027E/28E P/E, vs. the current price trading at 18x 2027E.

Three key earnings drivers

Driver 1: Caterpillar (CAT) genset cooling is the clearest near-term earnings driver. AIDC power shortages are accelerating demand for CAT gensets, while Yinlun's Mexico production, cost advantage and multi-year qualification status position it to capture incremental cooling-package share. We forecast genset revenue to rise from Rmb0.5bn in 2026E to Rmb3.2bn by 2028E (>140% CAGR), contributing >10% of Yinlun's revenue and ~20% of group profit by 2028E.

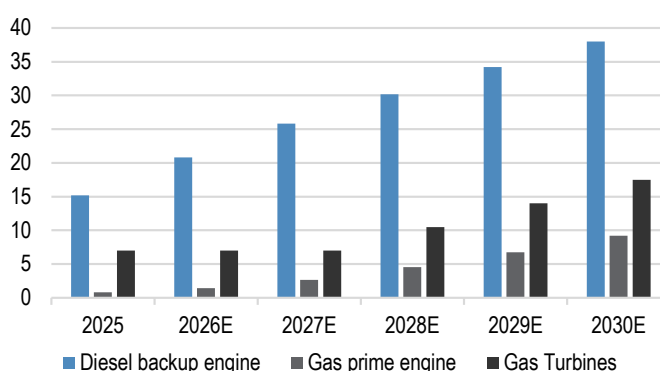
Figure 1: Genset revenue estimates and GP contribution to group

2026-28e revenue CAGR 143%



Source: J.P. Morgan estimates.

Figure 2: Caterpillar's capacity forecasts (by JPM US analyst Tami Zakaria)

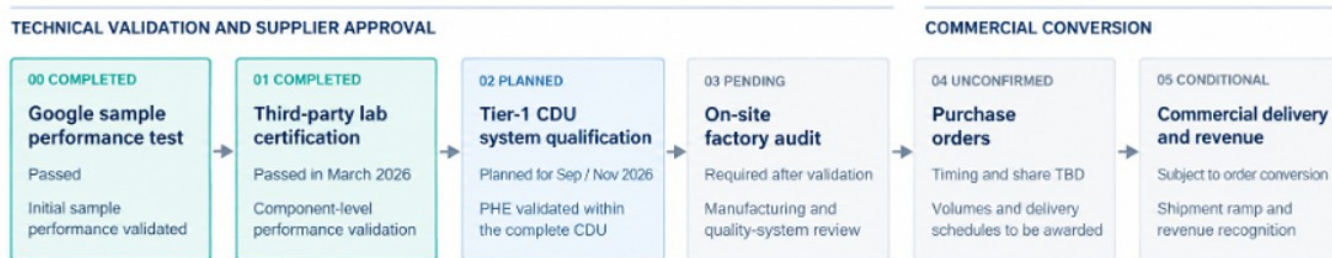


Note: Yinlun supplies cooling packages to Caterpillar's diesel backup engine and gas prime engine, not for gas turbines.

Source: Company data, J.P. Morgan estimates.

Driver 2: Liquid cooling creates a second AIDC growth leg, with potential Google orders as upside to our base case estimates. High-power PHEs are a core CDU component, while Yinlun's vacuum-brazing know-how, self-designed furnace capability and Thailand capacity plan support its entry into the global supply chain. Our base case excludes Google-related orders. Successful qualification and order conversion could provide 4-6% upside to our 2027-28E earnings estimates, with liquid-cooling profit contribution potentially approaching c.10% of group earnings by 2028E.

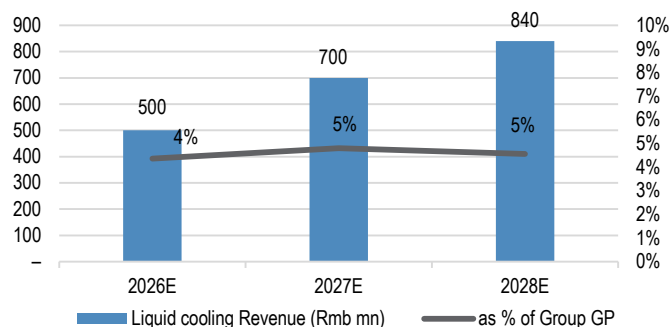
Figure 3: Google qualification progress



Source: Company data, J.P. Morgan.

Figure 4: Liquid cooling revenue estimates and GP contribution to group (base case)

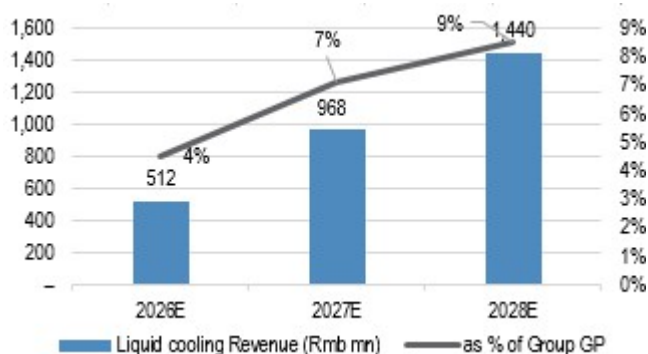
2026-28e revenue CAGR 30%



Source: J.P. Morgan estimates.

Figure 5: Liquid cooling revenue estimates and GP contribution to group (upside scenario)

2026-28e revenue CAGR 68%



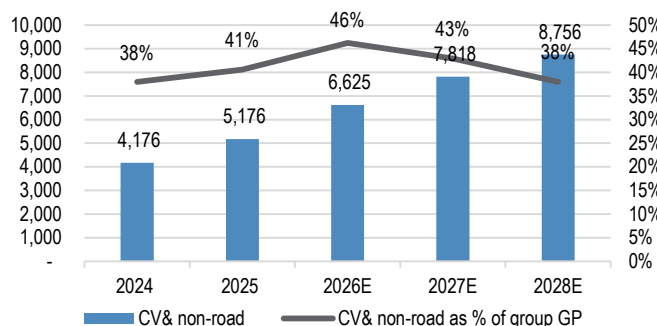
Source: J.P. Morgan estimates.

Driver 3: CV and non-road provide a resilient and higher-margin earnings base.

Export-led CV demand, domestic HDT replacement and accelerating HDT electrification should support c.20% revenue CAGR over 2025-28E. Yinlun is positioned to benefit both from the volume cycle and rising thermal-management content per vehicle as electric HDTs gain penetration. We expect HDT EV penetration to increase from ~25% in 2025 to ~50% by 2030E, supporting both revenue growth and content expansion.

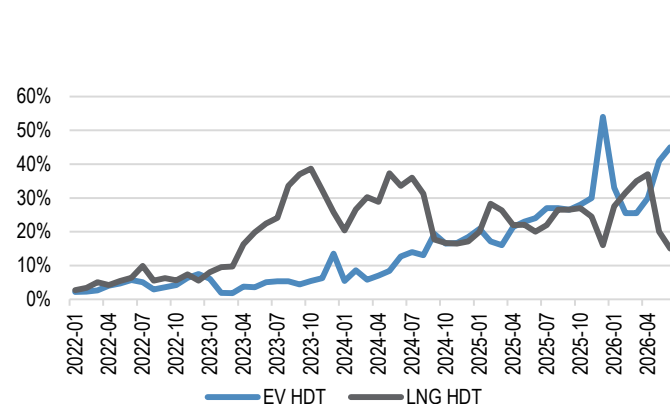
Figure 6: CV and non-road revenue estimates and GP contribution to group

2025-28e revenue CAGR 19%



Source: Company data, J.P. Morgan estimates.

Figure 7: China EV HDT vs. LNG HDT penetration



Source: CAAM.

Investment Thesis

1) AIDC-driven power demand: Caterpillar orders support growth visibility

AI-driven power demand is driving a multi-year genset upcycle

The rapid build-out of AI data centers is increasingly constrained by grid availability, driving demand for both backup-power diesel engines and prime-power gas engines. On its 2Q26 earnings call, Caterpillar highlighted strong AI- and data center-related power-generation demand, noting that customers are placing orders as far out as 2030. Diesel standby orders extend beyond 2028, gas-engine and gas prime-power lead times have reached late 2028, while gas-turbine delivery slots stretch into 2029. J.P. Morgan's US analyst Tami Zakaria forecasts Caterpillar's Power Energy revenue to grow at a 13% CAGR over 2025-30E and estimates AIDC backup-power diesel-engine capacity to increase by more than 2.5x from 2025 to 2030, while prime-power gas-engine capacity is expected to increase nearly ninefold over the same period. Yinlun supplies components for CAT's diesel and gas engines (not gas turbines).

Rising Caterpillar orders and high qualification barriers support revenue visibility

Yinlun began supplying generator cooling packages to Caterpillar in volume during 2H24. According to management, Caterpillar's qualification process for new suppliers typically takes two to three years, including performance and durability testing, factory audits and customer validation. These lengthy qualification cycles create meaningful barriers to entry and support incumbent suppliers' market positions.

According to Yinlun, Caterpillar has increased its 2026 cooling-package volume order to Yinlun to approximately 3,400 units, up 70% from around 2,000 units at the beginning of the year, while indicating approximately 5,000 units for 2027. At an ASP of roughly US\$20k for core products and up to US\$50k for newer products, these volumes imply cooling-package revenue of approximately US\$70mn in 2026 and US\$100mn in 2027. Notably, cooling-package net margins of 20-30% are materially above Yinlun's traditional automotive components business, supporting a more favorable earnings mix as volumes scale.

Beyond cooling packages, gas-engine aftertreatment represents an additional growth driver. Deliveries to Caterpillar are expected to begin in 4Q26 and reach 700-800 units in 2027 according to Yinlun, with an ASP of approximately US\$130,000 per unit, more than six times that of a standard cooling package.

Modine's strategic shift creates an opportunity for Yinlun to gain share

While end-market demand is accelerating, the competitive environment is becoming increasingly favorable for Yinlun. Modine (MOD US), historically Caterpillar's largest cooling-package supplier, is allocating an increasing share of capital and management attention toward data center liquid cooling rather than conventional genset thermal management.

In FY26 (ended March 2026), Modine's heavy-duty equipment revenue declined 5%, which management primarily attributed to lower genset revenue. Although the company did not disclose specific reasons, we believe the decline may partially reflect share gains by Yinlun at Caterpillar. More importantly, Modine's incremental US\$100mn capex

program, its more than US\$4bn data center capacity-locking agreements for 2027-29, and management attention are increasingly focused on North American AIDC cooling products such as chillers, AHUs and CDUs, rather than the traditional cooling package for genset.

The company has also announced plans to separate its legacy Performance Technologies operations, including heavy-duty equipment and genset cooling, through a combination with Gentherm. Following the transaction, Modine will focus capital allocation and strategic resources on Data Centers and Commercial HVAC (heating, ventilation and air conditioning), further reinforcing the company's shift away from traditional vehicle and genset thermal-management markets.

Meanwhile, Caterpillar continues to prioritize engine manufacturing and engineering resources while outsourcing selected aftertreatment and ancillary systems. Against a backdrop of rapidly rising genset demand and the need for supply-chain expansion, we believe Modine's relatively limited disclosed investment in genset capacity creates an attractive opportunity for Yinlun to secure incremental orders and increase content share.

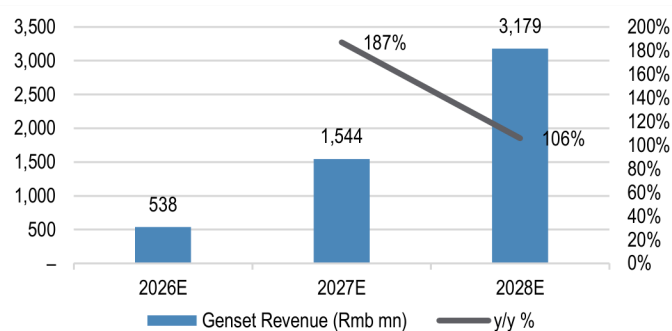
Cost advantages and localization support further share gains

With pricing approximately 20-30% below Modine, localized manufacturing in Mexico and established large-scale heat-exchanger production capabilities, Yinlun is well positioned to capture a larger portion of Caterpillar's growing outsourced component demand. We estimate Yinlun's share of CAT's cooling-package requirements could increase by ~10ppt to more than 50% by 2028E.

A meaningful earnings contributor by 2028E

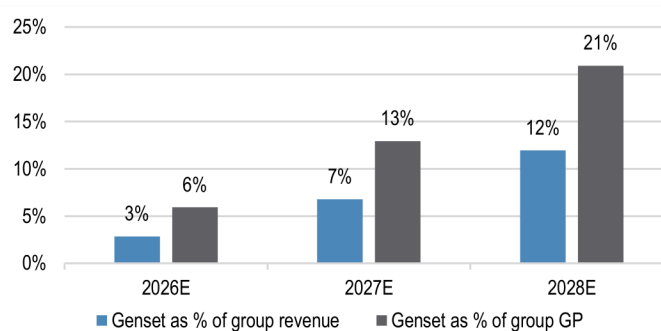
We forecast genset-related revenue to increase from Rmb0.5bn in 2026E to Rmb1.5bn in 2027E and Rmb3.2bn in 2028E, representing approximately 3%, 7% and 12% of group revenue, respectively. Given the business's above-average profitability, we expect gensets to contribute 6%, 13% and 21% of group gross profit in 2026-28E, positioning AIDC-driven power demand as one of Yinlun's most important earnings growth drivers over the next three years.

Figure 8: Genset revenue estimates and yoy%



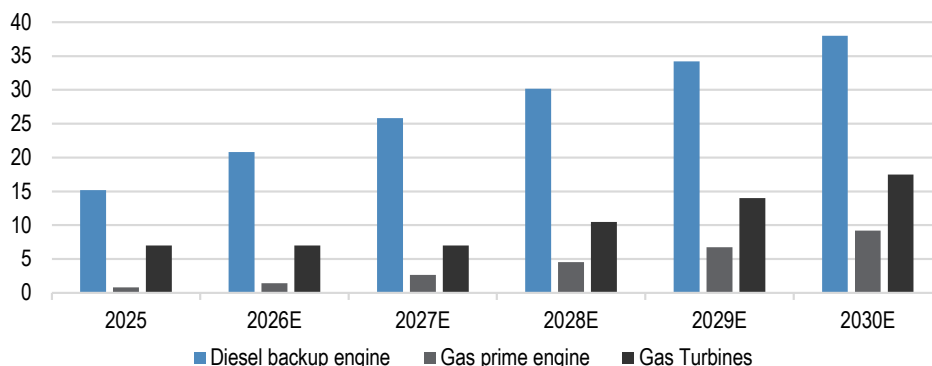
Source: Company data, J.P. Morgan estimates.

Figure 9: Genset revenue and GP contribution



Source: Company data, J.P. Morgan estimates.

Figure 10: CAT's capacity forecasts (by JPM US analyst Tami Zakaria)



Note: Yinlun supplies cooling packages to Caterpillar's diesel backup engine and gas prime engine, not for gas turbines.
Source: Company data, J.P. Morgan estimates.

Table 1: Yinlun's genset-related product supply to Caterpillar (CAT)

Product	Prior supplier / supply model	Yinlun entry timing	Volume expectation (units)	ASP assumption
cooling package	Primarily Modine	Began volume supply in 2H24	JPM 3,700 / 5,500 / 8,250 units in 2026E / 2027E / 2028E	~US\$20k per unit
aftertreatment system	Previously manufactured in-house by CAT	Initial deliveries expected in 4Q26	JPM 20 / 800 / 2,000 units in 2026E / 2027E / 2028E	~US\$130k per unit

Source: Company data, J.P. Morgan estimates.

2) Liquid cooling creates a second AI data center growth opportunity; Google orders are upside optionality

PHEs account for 10-20% of CDU costs, offering meaningful content per unit

Plate heat exchangers (PHE) represent one of the largest components of a CDU's BOM, accounting for approximately 10–20% of CDU cost, behind circulation pumps at 30–50%. Each CDU incorporates around three PHEs at ASP of approximately RMB40k-50k each, implying revenue content of roughly RMB120k-150k per CDU. This positions Yinlun to benefit from liquid-cooling adoption as a core component supplier, with revenue growth supported by rising downstream CDU shipments.

Vacuum brazing expertise and in-house furnace capability underpin Yinlun's scale-up in PHEs

- Vacuum brazing: the yield-critical bottleneck in high-power BPHE capacity.**
 High-power CDU-embedded brazed plate heat exchanger (BPHE) requires compact designs that deliver high heat-transfer efficiency, low pressure drop, and long-term sealing reliability, placing high demands on flow-channel design, plate forming, and vacuum brazing. Brazing quality — governed by temperature uniformity, vacuum level, and process curve — directly determines weld integrity and leak risk, so stable mass production depends on a tight match between equipment and process. This makes the vacuum brazing furnace both the key equipment for yield and the key bottleneck for capacity: its parameters must be custom-built per product, so capacity cannot be quickly replicated with off-the-shelf equipment.
- Furnace supply constraints cap the pace of overseas capacity expansion:**
 Overseas leaders such as Alfa Laval (ALFA SS Equity) and SWEP (subsidiary of

Dover, DOV US Equity) source their high-end brazing furnaces largely from European equipment makers such as TAV (private company) and Ipsen Global (private company), which themselves face capacity and scheduling constraints — customized furnace orders typically carry lead times exceeding six months, and once installation, commissioning, and yield ramp-up are added, overall expansion cycles generally run 2–3 years.

- **Four decades of manufacturing experience underpin in-house furnace design:** Yinlun has produced plate heat exchangers since 1980, accumulating over 40 years of manufacturing experience. Its vacuum brazing furnaces are self-designed and outsourced for fabrication, allowing equipment parameters to be optimized around product requirements and creating synergy between process know-how and equipment design. Yinlun's competitiveness rests not only on its manufacturing heritage but also on this in-house design capability, which reduces reliance on external furnace suppliers' production schedules and gives the company greater initiative in product iteration and capacity expansion — laying the groundwork to capture incremental demand for PHEs.
- **Capacity expansion tied to localization, targeting global footprint:** Yinlun has tied its capacity expansion to localization: its first large-scale PHE line (15k units/year) will come online in 4Q26, with two more planned for 2027 — one in South China, one in Thailand — forming a "2 domestic + 1 Thailand" layout aimed at global localized supply. At 15k units/line and an ASP of ~RMB 40k, the three lines together imply ~RMB 1.8bn in annualized output value.

Figure 11: Overseas PHE players' capacity expansion

Company	Prior Expansion	Latest Round
Alfa Laval	May 2023: Announced capacity expansion across four plants — San Bonifacio (Italy), Jiangyin (China), Ronneby (Sweden), and Richmond (USA) — for brazed plate heat exchangers	Plans to double capacity at its Virginia (Richmond) plant by 2028; added RMB 1bn investment in its Jiangyin, China plant in June 2026
SWEP	Launched its "Global Investment Program" in 2019 In October 2022 , announced expansion of its Suzhou plant into a larger new facility (occupied in early 2024), plus a new R&D center in Landskrona	June 2026: Committed an additional \$30M investment to more than double large-size product capacity by 2026–2027 across all five global sites
Kelvion	September 2023: Expanded its Sarstedt, Germany plant, adding 150,000 units/year of capacity (this facility is known to produce gasketed plate heat exchangers (GPHE) based on case studies; unclear whether brazed units are included)	August 2025: Added ~55,000 sqm across Tulsa, US (42,000 sqm, making it Kelvion's largest global manufacturing site) and Opole, Poland — explicitly tied to "data center cooling" demand, though not explicitly labeled "brazed"
Danfoss	n/a	Building a new BPHE production line in Monterrey, Mexico, with two additional brazing furnaces planned by end-2026 (the only confirmed BPHE-specific capacity expansion announcement identified for Danfoss to date)

Source: Company data.

Track record in PHEs supports Yinlun's ability to take share from established overseas suppliers

Yinlun has manufactured stainless-steel plate heat exchangers since 1980, building more than four decades of experience in plate forming, vacuum brazing and mass-production quality control. In commercial-vehicle engine applications, Yinlun has established an estimated global market share of around 95% in stainless PHEs and close to 100% in China, reflecting its long-standing technical position in a demanding engine-side application. Yinlun has also replicated this capability in HVAC heat-pump applications. Alfa Laval previously held a leading position, but Yinlun began taking share in 2022–23,

supported by AlfaLaval's capacity constraints and Yinlun's price competitiveness. Yinlun has reached an estimated 70-80% market share in China and a meaningful position in Europe, supplying customers including Gree, Midea and Hisense for commercial-building applications.

Yinlun in Google's qualification process – upside optionality

Data center PHE procurement follows a tiered gatekeeping structure, under which CDU integrators handle component selection while hyperscalers retain effective veto power over supplier qualification. Entry into Google's supply chain follows a sequential qualification process. Yinlun first completed a Google-run sample performance test. Following a successful result, Google issued a formal confirmation and referred the product to its designated third-party laboratory for further certification. After the lab certification is completed, the result is communicated to the relevant CDU integrator, which then validates Yinlun's PHE within its CDU module.

Yinlun has passed Google's initial sample performance test and subsequently cleared the Google-designated third-party laboratory certification in March 2026. The next gate is CDU module validation. Tier-1 CDU integrator qualification — covering partners such as Luxshare, Kehua and Deltaetc— is scheduled to advance through September and November. If any one CDU integrator completes module validation successfully, Yinlun could be invited for a Google factory audit in China, potentially as early as November, before progressing to purchase-order approval. If completed successfully, this would open a new order channel for the PHE business; at this stage, however, it should still be viewed as potential upside rather than confirmed revenue, with actual monetization contingent on order conversion.

Figure 12: Google's qualification process

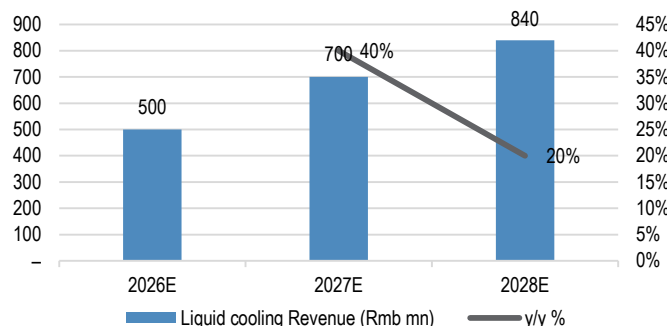


Source: Company data, J.P. Morgan.

4-6% potential EPS upside if Google orders materialize

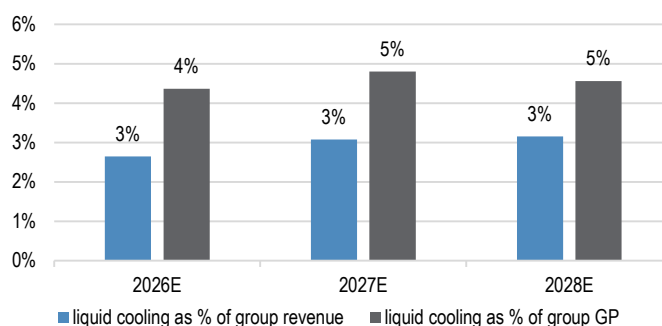
Our base-case estimates assume no Google-related orders and forecasts liquid-cooling revenue to grow from Rmb0.5bn in 2026E to Rmb0.8bn in 2028E, equivalent to approximately 3% of group revenue and 4-5% of group gross profit over the period. Successful Google qualification and order conversion could increase liquid-cooling revenue to approximately Rmb1.4bn by 2028E, lifting segment contribution to around 5% of group revenue and 9% of group gross profit. We estimate this would translate into net profit upside of approximately 4% / 6% in 2027E / 2028E, respectively, relative to our base case forecasts.

Figure 13: Liquid cooling revenue estimates and yoy% (base case)



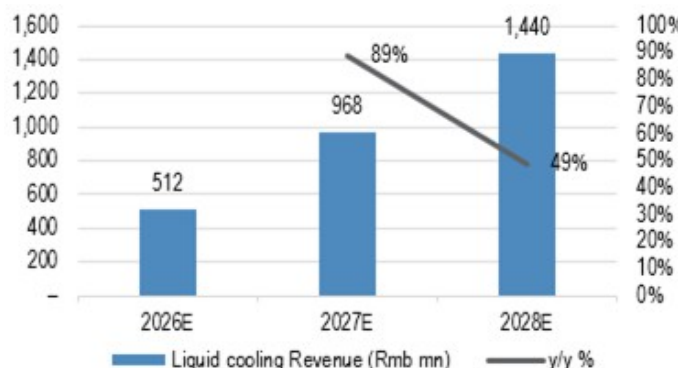
Source: J.P. Morgan estimates.

Figure 14: Liquid cooling revenue and GP contribution (base case)



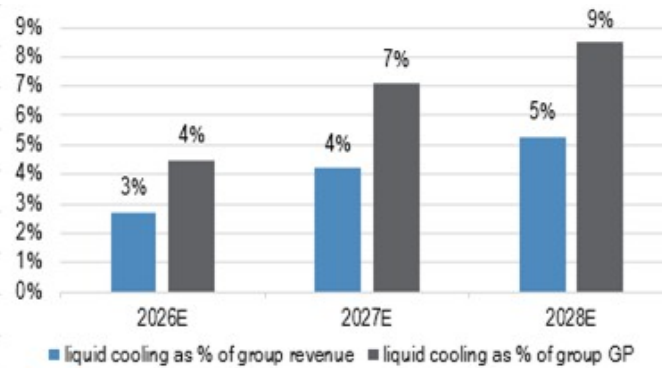
Source: J.P. Morgan estimates.

Figure 15: Liquid cooling revenue estimates and yoy% (upside scenario)



Source: J.P. Morgan estimates.

Figure 16: Liquid cooling revenue and GP contribution (upside scenario)



Source: J.P. Morgan estimates.

3) Commercial vehicle electrification and exports provide a resilient growth foundation

Multiple demand drivers point to a steady, export-led up-cycle into 2027

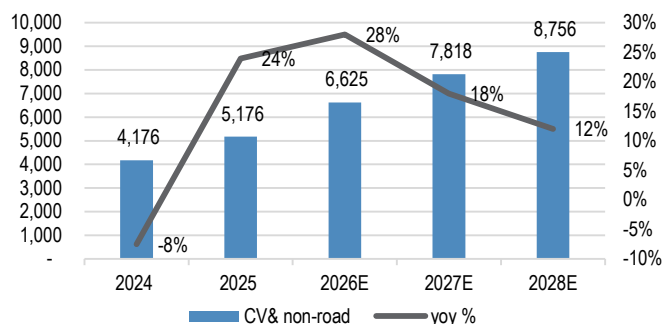
Looking into 2027, we expect exports to remain the primary driver of CV volume growth, with domestic demand supported by a steady HDT replacement cycle.

- **Exports:** Sustained infrastructure buildout and resource extraction activity across Africa, Southeast Asia, and Latin America continue to underpin overseas demand, while Chinese CV makers' cost-performance advantage and expanding overseas KD assembly footprints continue to take share from legacy European and American brands — supporting continued decent export growth.
- **Domestic HDT replacement demand in 2027 should be underpinned by two complementary cohorts.** China IV trucks, mostly registered in 2015–2017, will be 10–12 years old by 2027 and are already entering a high-scrappage phase; with driving restrictions and rising maintenance costs adding further pressure, this cohort's accelerated retirements driven by this year's policy support are expected to bring further replacement demand next year. China V trucks, registered mostly in

2017–2020 during the last HDT upcycle, will reach 7–10 years of age in 2027 and are also starting to enter their replacement cycle, which should contribute a degree of incremental replacement demand next year. Together, these two cohorts point to a steady replacement cycle for domestic HDT demand in 2027.

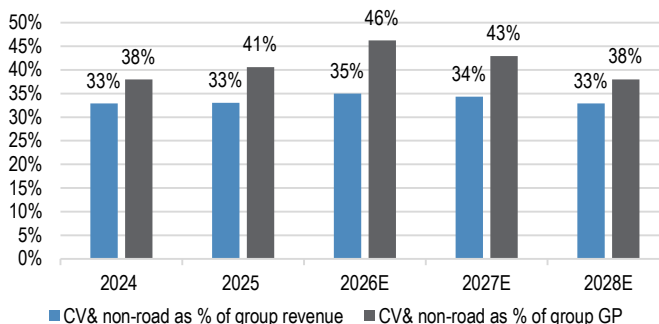
Yinlun is well positioned to capture the export-driven CV up-cycle and HDT electrification trend. Commercial vehicle & non-road is Yinlun's second-largest segment, contributing ~37% of 1H26 revenue (Rmb3.32bn, +37% y/y) and ~48% of gross profit (Rmb785mn, at a ~24% gross margin — above the company's average of ~18% and well above the passenger vehicle segment's ~13% margin). ~30% of Yinlun's commercial vehicle & non-road revenue comes from overseas markets. We forecast the segment to grow 28%/18% y/y in FY26E/FY27E, driven by a steady export-led up-cycle, together with the domestic HDT replacement cycle and HDT electrification trend.

Figure 17: CV and non-road revenue estimates (Rmb mn) and yoy%



Source: Company data, J.P. Morgan estimates.

Figure 18: CV and non-road revenue and GP contribution



Source: Company data, J.P. Morgan estimates.

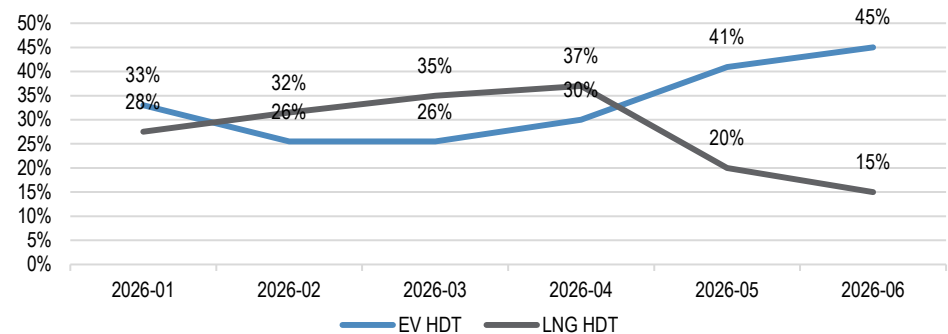
HDT electrification accelerating, backed by a new national policy roadmap

HDTs are now a meaningful swing factor within the broader CV market — accounting for ~28% of China's CV wholesales in 8M26 per CAAM data.

On 12 June 2026, China released its first national Master Plan for HDT electrification, setting a 40% new-energy penetration target by 2030 — equivalent to roughly 1.6mn new-energy HDTs, or ~20% of the installed HDT parc. The plan also calls for 3,000 new HDT charging/battery-swap stations, 80% electrification of fixed-route short-haul freight in key regions, and 30,000km of zero-carbon freight corridors, providing a clearer policy runway for adoption ([JPM report](#)).

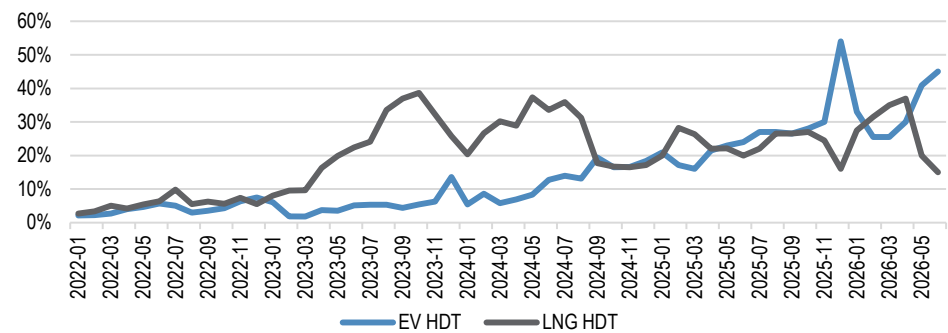
We see the 40% target as a floor rather than a ceiling: China has a track record of overachieving similar policy goals (e.g., the 2020 passenger NEV plan targeted 20% penetration by 2025, which was surpassed), and EV HDT penetration has already tracked ahead of a linear path to the target — rising from ~5% in 2023 to ~25% in 2025 and ~33% YTD in 1H26 (with Jun alone reaching ~45%), aided by improving battery technology, expanding charging/swap infrastructure, and rising LNG price volatility that strengthens the TCO case for EVs. We therefore see EV HDT penetration reaching roughly 50% by 2030E, above the government's stated target.

Figure 19: China EV HDT vs. LNG HDT monthly penetration in YTD26



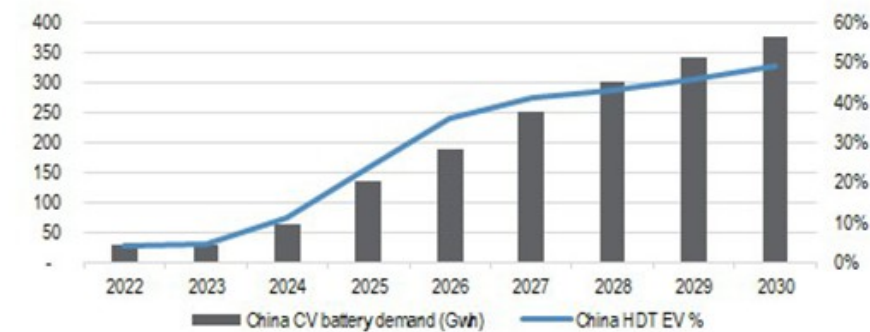
Source: CAAM.

Figure 20: China EV HDT vs. LNG HDT penetration



Source: CAAM.

Figure 21: China CV battery demand and HDT NEV % forecast



Source: GGII, CAAM, J.P. Morgan estimates.

Figure 22: HDT snapshot

	2024	2025	1H26
China HDT sales (source: CAAM)			
Total HDT wholesales (incl exports) (k units)	902	1,144	661
Total domestic HDT sales (k units)	611	675	410
NEV HDT sales (k units)	69	192	135
<i>EV penetration in domestic HDT</i>	<i>11%</i>	<i>28%</i>	<i>33%</i>
China domestic HDT sales(source: GGII)			
NEV HDT sales (k units)	n/a	224	139
Total NEV truck sales (k units)	486	679	412
Total NEV CV (truck + bus) sales (k units)	535	730	427
<i>NEV HDT as % of total NEV CV</i>		<i>31%</i>	<i>33%</i>

Source: CAAM, GGII.

Electrification drives content per car higher, a structural tailwind for Yinlun

As electrification penetrates deeper into HDT, thermal-management requirements shift from radiators, intercoolers, oil coolers, and EGR coolers built around the engine and exhaust aftertreatment system, toward battery thermal management, motor/power-electronics cooling, and cabin heating and cooling. In the conventional diesel HDT, these components carry content per vehicle of roughly RMB14-15k; once the engine is replaced, these ICE-specific parts are no longer used, replaced instead by three-electric thermal-management parts — battery cooling plates, battery chillers, motor/e-drive oil coolers, high-voltage PTC heaters, electronic water pumps, and electronic water valves — lifting content per vehicle to roughly RMB20-30k, roughly double the conventional HDT.

Yinlun's product portfolio is well positioned to capture this uplift, spanning battery thermal-management systems (BTMS), PTC heaters, coolant valves, and HVAC modules alongside its established conventional cooling products.

80%-90% of Yinlun's CV business remains ICE-based. Amid HDT electrification, if core customers such as FAW Jiefang, Weichai etc scale up their electric truck offerings, Yinlun's 20+ year deep partnership puts it in a strong position to win supply deals and increase electric-related revenue and per-unit content in its CV segment. As electric HDT adoption rises, this shift in content mix and higher per-vehicle value should drive Yinlun's commercial vehicle business to grow faster than underlying unit volumes.

Investment Risks

1) High customer concentration in genset business creates significant dependence on Caterpillar

Caterpillar has become increasingly important to Yinlun's growth outlook

Caterpillar accounted for c.16% of Yinlun's commercial-vehicle and non-road revenue, or c.6% of group revenue, in 2025. As genset cooling-package revenue ramps, we expect CAT's contribution to Yinlun's consolidated revenue to increase meaningfully, further raising the company's exposure to CAT's order cadence, sourcing decisions and pricing negotiations.

We forecast genset-related revenue to increase from Rmb0.5bn in 2026 to Rmb1.5bn in 2027 and Rmb3.2bn in 2028, representing 3%, 7% and 12% of group revenue as well as 6%, 13% and 21% of group level GP, respectively.

While Yinlun is developing other genset customers, near-term growth remains heavily dependent on Caterpillar. Changes in Caterpillar's order schedule, production plan or inventory policy could therefore have a disproportionate impact on earnings delivery.

Qualification barriers provide some protection but do not secure long-term share

Yinlun currently supplies approximately half of Caterpillar's cooling-package requirements, while CAT has begun outsourcing part of the aftertreatment production previously handled in-house. However, Caterpillar retains control over sourcing decisions and could increase internal production, qualify additional suppliers or reallocate volumes among existing vendors. The lengthy qualification process reduces near-term substitution risk but does not eliminate longer-term competition, pricing pressure or potential share loss.

2) Liquid-cooling PHE orders have yet to be secured, while capacity is being added ahead of confirmed demand

Small-batch orders and ongoing validation do not yet establish Yinlun as a primary supplier

Yinlun's PHEs for Google-related applications remain at the CDU system-validation stage, with no confirmed volume orders to date. The company is still largely positioned as an incremental or secondary supplier, with current opportunities partly driven by order overflow from established global vendors such as Alfa Laval. Entry into a hyperscaler supply chain requires approval by the CDU integrator as well as component-level qualification by the cloud customer. For Google, Yinlun passed third-party laboratory testing in March 2026 but must still complete Tier-1 CDU system validation, an on-site factory audit and purchase-order approval. Delays or unsuccessful results at any stage could defer revenue conversion. Even after qualification, purchasing volumes will depend on end demand, incumbent suppliers' available capacity and Yinlun's ability to deliver consistently at scale.

Capacity additions ahead of order visibility create utilization and margin risks

Yinlun's first production line, with annual capacity of 15,000 large-format PHEs, is scheduled to commence operations in 4Q26. The company also plans to add lines in South China and Thailand in 2027, bringing the potential total to three. At an ASP of

approximately Rmb40,000 per unit, the three lines could support up to Rmb1.8bn of annual revenue at full utilization; this represents potential capacity rather than secured orders. If customer qualification or order conversion takes longer than expected, upfront investment, depreciation, staffing costs and initial yield ramp-up could weigh on utilization and profitability.

3) Geopolitical risk

US “de-risking” policies have recently expanded from advanced computing into power infrastructure, increasing concerns over localized sourcing in the AIDC supply chain

On the one hand, the US Bureau of Industry and Security (BIS) continues to tighten controls on exports to China of advanced computing chips, HBM, semiconductor equipment and AI/supercomputing end uses ([news](#)). It will also reinstate the Affiliates Rule in November 2026, extending restrictions to overseas affiliates that are 50%-or-more owned by entities on the Entity List and increasing supply chain due diligence requirements ([news](#)). On the other hand, President Trump signed EO 14421 in August 2026, bringing transformers, energy storage systems, grid-connected inverters, generators, gas turbines and industrial control systems within the scope of US bulk power system security review, with DOE required to formulate implementing rules within 120 days ([news](#)). Although these measures do not directly restrict Chinese thermal-management components, they point to a broader policy direction of de-risking the computing and power infrastructure underpinning AI data centers, raising market concerns over localized procurement across the AIDC supply chain.

We believe the direct policy risk to Yinlun’s AIDC liquid cooling components and genset cooling packages remains limited.

- PHEs and other CDU thermal-management components are not among the chips, servers, power electronics equipment or control systems currently targeted by export controls or EO 14421. Purchasing decisions should therefore continue to be driven primarily by performance, reliability, lead time and cost. As AIDC deployment scales, hyperscalers and Tier-1 CDU integrators also need to manage system costs, preserving a role for cost-competitive Chinese suppliers of non-sensitive and relatively standardized components. Yinlun is also building PHE capacity in Thailand, providing a regional supply option for overseas customers.
- In terms of genset cooling packages, the EO 14421 order does not specifically identify genset cooling packages or other thermal-management components as covered equipment; its focus is on generators, turbines, grid-connected power equipment and industrial control systems. We believe Yinlun’s genset cooling packages are therefore not currently a direct target of the policy, while Mexico-based production further supports localized delivery to CAT.

Share Price Analysis

2023-24: NEV passenger vehicle thermal management drove core growth

In 2023, Yinlun's share price significantly outperformed the CSI300 (Yinlun +50% vs. CSI300 -11%), driven by strong NEV passenger vehicle sales growth in China, rising market share with core OEM customers, and higher per-vehicle thermal management content from electrification. Passenger vehicle thermal management revenue grew 51% YoY in 2023, becoming the primary driver of both revenue and earnings growth.

In 2024, passenger vehicle revenue continued to grow at 31% YoY, but this was dragged down by an 8% YoY decline in the commercial vehicle segment, as well as intensifying price competition across the domestic auto supply chain and market caution on legacy thermal management valuations. The stock rose only modestly (0.3% vs. CSI300 +15%), underperforming the index but still outperforming most of the china auto parts names, which fell sharply for the year thanks to the 43% growth of the new business digitakl and energy segment, which fell sharply for the year, thanks to the 47% growth of the new Digital and Energy segment.

2025: Commercial vehicle recovery supported earnings, while the AIDC opportunity drove a valuation re-rating

In 2025, Yinlun's share price sharply outperformed the CSI300 (+102% vs. CSI300 +18%), supported by a recovery in commercial vehicle and non-road demand (replacement cycle, exports, and non-road applications), with commercial vehicle and non-road revenue up 24% YoY — the fundamental earnings driver for the year. At the same time, the market's attention increasingly shifted to the company's entry into Caterpillar genset cooling packages, the North American AIDC power shortage opportunity, and the long-term growth potential of liquid cooling heat exchangers. We view the 2025 outperformance as driven jointly by commercial vehicle earnings delivery plus upward revisions to AIDC expectations.

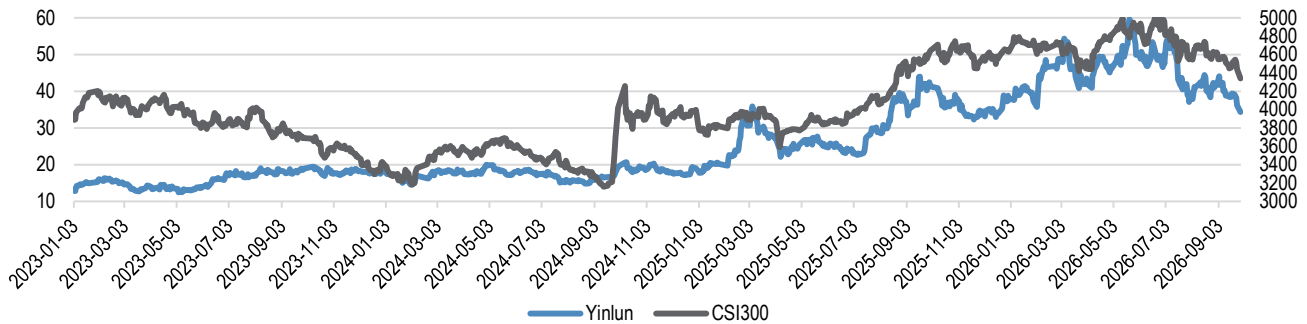
2026 YTD: Shift from thematic narrative to order/earnings verification

As of 29-Sep 2026, Yinlun's share price is down ~10% from end-2025 levels, underperforming CSI300 (CSI300 -6%).

- Early in the year, the market rapidly revised up expectations around the North American AIDC power shortage, Caterpillar's genset capacity expansion, and Yinlun's entry into the cooling package supply chain; upgraded order guidance for the genset business and the gradual ramp-up of overseas local capacity supported the stock in phases.
- Since then, however, market pricing logic has shifted from "theme and TAM" to "orders, capacity, and earnings delivery." On one hand, while the genset business has high growth visibility, incremental volume still depends heavily on Caterpillar, and the market needs continued confirmation of actual procurement cadence, supply share, and the ramp of higher-margin products. On the other hand, liquid cooling heat exchangers remain in the customer qualification, Tier-1 integration validation, and small-batch order stage — progress with leading customers such as Google has not yet translated into confirmed large-scale orders, so the long-term addressable market has not yet been fully reflected in near-term revenue.
- Fundamentals reflect this transitional phase: in 1H26, revenue growth remained

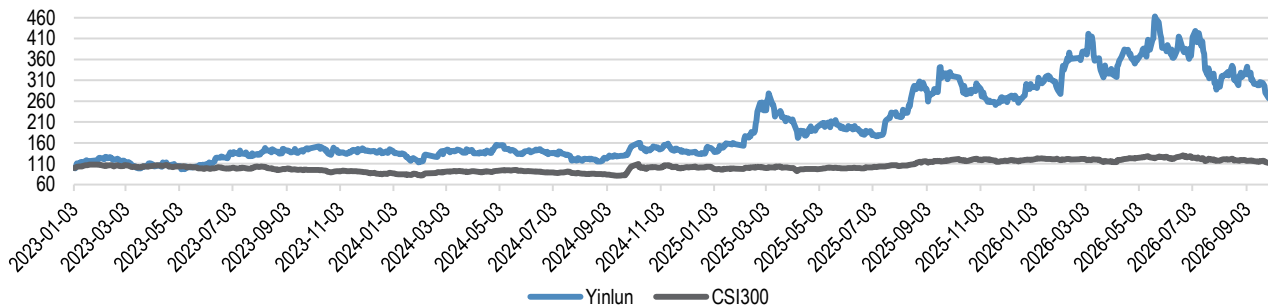
solid, but net profit grew more slowly than revenue, reflecting pricing pressure in legacy passenger vehicle thermal management. We believe 2026 YTD performance does not signal a weakening of the AIDC thesis, but rather reflects a pause after the 2025 re-rating, with the market awaiting further confirmation on liquid cooling orders.

Figure 23: Yinlun share price vs. CSI300



Source: Bloomberg Finance L.P.

Figure 24: Yinlun share price vs. CSI300 (index 3 Jan 2023 price to 100)



Source: Bloomberg Finance L.P.

Valuation

We value Yinlun using a sum-of-the-parts (SOTP) approach, given the increasingly divergent growth and margin profiles across its traditional business lines – commercial and passenger vehicle auto parts – vs. new business lines such as genset cooling packages and AIDC liquid cooling.

We forecast group revenue to grow at a 15% CAGR over 2026-30, but this masks a sharp mix shift toward the Digital and Energy segment (includes genset, liquid cooling and ESS thermal control parts), which we expect to grow at a >50% CAGR over the same period – with genset cooling packages and liquid cooling individually growing at >60% CAGRs, as the company rides the AIDC capex cycle and related growth in prime and backup power demand. Both genset and liquid cooling carry gross margins structurally well above the legacy auto parts business, so the rising revenue contribution from these segments (CV and PV parts falling from 90% of NP in 2026 to 75% by 2028, based on JPMe) drives margin expansion at the group level, with net profit growth outpacing revenue growth in 2027-28 as the higher-margin segments scale.

Table 2: Revenue target in 2030 vs. 2026 JPMe

	JPMe 2026E Revenue (Rmb mn)	Revenue target (Rmb mn) (company guidance)	as % of group level revenue	2026-30E CAGR
Group level revenue	18,921	33,000	100%	15%
Digital and energy	1,938	10,000	30%	51%
Genset	538	4,000	12%	65%
Liquid cooling	500	4,000	12%	68%
ESS	900	2,000	6%	22%

Source: Company data, J.P. Morgan estimates.

Our Dec-27 price target of Rmb51 does not include a contribution from potential Google-related liquid cooling orders. Given the differentiated growth and margin trajectories, we apply segment-specific 2028E P/E multiples rather than a blended group multiple: 15x for PV and CV parts, in line with mature auto-parts peers, and 35x for the Digital & Energy business, reflecting premium growth-stage multiples for AIDC power infrastructure and liquid cooling exposure. Applying these multiples to our 2028E segment earnings forecasts results in a base case SOTP value of Rmb51, implying ~49% upside to the current share price. This implies a blended 29x 2027E P/E or 20x 2028E P/E.

Our upside case assumes successful conversion of Google-related PHE qualification into commercial orders. Under this scenario, higher liquid cooling revenue lifts Digital & Energy earnings, while the valuation framework remains unchanged. This results in an SOTP-based Dec-27 valuation of Rmb55, implying ~61% upside to the current share price. We view this as potential upside rather than part of our base case, as Google-related orders remain subject to tier-1 CDU validation, factory audit and purchase-order conversion. This implies a blended 31x 2027E P/E or 22x 2028E P/E.

We believe this SOTP framework better captures the value of Yinlun's optionality in the AIDC power infrastructure and cooling theme than a single blended multiple.

Table 3: Yinlun SOTP valuation (base case)

Segment	2028E EPS (Rmb)	P/E multiple	PT (Rmb)
PV and CV parts	1.9	15	28.2
Digital and energy	0.6	35	22.5
Total			51
Implied 2028E P/E			20

Note: Base case earnings estimates do not include contribution from potential Google-related liquid-cooling orders.
Source: J.P. Morgan estimates.

Table 4: Yinlun SOTP valuation (upside case)

Segment	2028E EPS (Rmb)	P/E multiple	PT (Rmb)
PV and CV parts	1.9	15	28.7
Digital and energy	0.8	35	26.7
Total			55
Implied 2028E P/E			22

Note: Upside case earnings estimates include contribution from potential Google-related liquid-cooling orders.
Source: J.P. Morgan estimates.

Table 5: Valuation comparison

Date: 20260929														
Company	BBG ticker	Rating	Price local currency	Mkt cap (USD mn)	2026E P/E	2027E P/E	2028E P/E	2026E net profit growth	2027E net profit growth	2028E net profit growth	2026E ROE	2027E ROE	2028E ROE	YTD price perf.
PV/CV thermal management parts														
Yinlun Machinery	002126 CH	OW	34	4,320	24.6x	17.7x	13.4x	19.3%	41.3%	31.6%	14.3%	17.0%	18.6%	(9.5%)
Sanhua Intelligent Controls	002050 CH	OW	36	21,321	33.7x	28.4x	24.4x	6.7%	18.6%	16.4%	13.1%	14.1%	14.8%	(35.5%)
Tuopu Group	601689 CH	UW	47	12,274	28.4x	23.0x	20.0x	(2.5%)	20.8%	19.0%	11.2%	12.6%	13.3%	(38.6%)
Zhongding Group	000887 CH	NC	18	3,623	14.6x	12.7x	10.9x	(2.9%)	15.4%	16.9%	10.6%	11.0%	11.5%	(20.5%)
Sichuan Chuanhuan Technology	300547 CH	NC	24	767	19.0x	15.9x	13.5x	15.1%	18.8%	18.0%	17.6%	18.1%	18.4%	(39.8%)
Chongqing Sulian Plastic	301397 CH	NC	22	667	31.6x	25.1x	21.1x	(4.9%)	24.8%	19.1%	6.8%	8.2%	9.2%	(17.4%)
Denso	6902 JP	UW	1847	34,154	11.7x	10.1x	9.2x	(3.6%)	14.3%	7.5%	7.4%	8.1%	8.6%	(14.4%)
Average					23.4x	19.0x	16.0x	3.9%	22.0%	18.4%	11.6%	12.7%	13.5%	(25.1%)
Median					24.6x	17.7x	13.5x	-2.5%	18.8%	18.0%	11.2%	12.6%	13.3%	(20.5%)
Genset OEM and components														
Caterpillar	CAT US	OW	820	376,910	30.2x	25.2x	21.0x	42.9%	16.8%	16.9%	64.3%	68.5%	65.0%	43.1%
Cummins	CMJ US	N	518	71,288	17.7x	15.0x	12.9x	27.4%	16.9%	15.0%	28.3%	31.8%	29.1%	1.4%
Weichai Power	000338 CH	OW	25	31,999	14.5x	10.9x	9.0x	20.5%	27.3%	27.8%	14.9%	17.9%	19.3%	45.3%
Jereh	002353 CH	OW	110	16,766	31.9x	15.5x	11.4x	14.1%	105.4%	36.7%	14.1%	24.2%	25.7%	55.0%
Generac	GNRC US	OW	205	12,109	21.1x	16.5x	13.3x	48.6%	28.5%	22.9%	17.4%	19.0%	19.8%	50.5%
Rolls-Royce	RRY LN	OW	1485	163,141	34.4x	30.1x	25.8x	47.7%	11.1%	14.6%	126.5%	131.2%	156.4%	29.1%
Wartsila	WRT1V FH	N	29	19,139	24.9x	21.9x	19.2x	7.2%	16.2%	14.9%	24.0%	26.2%	27.0%	(4.8%)
GE Vernova	GEV US	OW	950	252,956	39.4x	38.7x	27.3x	242.2%	(6.0%)	39.5%	51.1%	40.4%	43.2%	45.3%
Siemens Energy	ENR GR	OW	143	139,632	32.0x	22.8x	17.6x	167.5%	38.2%	28.4%	33.9%	41.6%	46.2%	18.7%
Mitsubishi Heavy Industries	7011 JP	NC	3793	81,287	29.1x	24.9x	21.4x	52.5%	17.2%	16.2%	13.2%	14.0%	14.2%	(1.2%)
Baker Hughes	BKR US	OW	57	56,702	22.3x	18.9x	15.8x	4.2%	16.6%	18.4%	13.2%	15.0%	16.7%	25.4%
Dongfang Electric	600875 CH	NC	23	11,396	16.4x	14.1x	12.0x	24.6%	18.3%	17.9%	10.2%	11.1%	12.1%	(5.4%)
Yingliu	603308 CH	OW	37	3,997	46.5x	29.2x	19.6x	35.1%	59.1%	48.2%	10.0%	14.3%	18.2%	(11.9%)
Wanze	000534 CH	NC	22	1,669	35.4x	24.5x	17.3x	30.1%	44.1%	41.7%	16.2%	18.5%	20.2%	(0.2%)
Howmet Aerospace	HWM US	OW	228	91,054	42.9x	35.0x	29.6x	43.0%	20.5%	17.4%	36.5%	37.2%	37.5%	11.4%
Woodward	WWD US	NC	326	19,245	34.4x	30.3x	25.7x	41.2%	12.6%	16.5%	22.5%	24.4%	27.1%	7.8%
Modine	MOD US	NC	175	9,297	23.3x	15.7x	11.3x	196.3%	52.1%	40.9%	27.1%	32.4%	33.1%	31.1%
Wabtec	WAB US	N	292	49,384	26.9x	23.5x	20.8x	19.9%	12.5%	11.1%	16.2%	17.6%	18.9%	37.0%
Sanhe Tongfei Refrigeration	300990 CH	NC	98	2,513	44.5x	26.4x	19.7x	39.4%	68.9%	34.1%	16.1%	22.8%	24.5%	9.2%
Average					30.2x	23.5x	19.2x	39.4%	18.3%	18.4%	29.3%	32.0%	34.4%	18.7%
Median					30.2x	23.5x	19.2x	39.4%	18.3%	18.4%	17.4%	24.2%	25.7%	18.7%
Liquid cooling integrators and components														
Vertiv	VRT US	NC	244	93,953	36.6x	26.7x	21.1x	64.5%	37.0%	28.0%	49.5%	47.5%	44.4%	50.6%
Schneider Electric	SU FP	OW	298	195,455	29.2x	24.5x	21.2x	17.1%	18.7%	15.6%	21.7%	23.2%	23.8%	26.7%
Delta Electronics	2308 TT	OW	1835	149,396	45.4x	28.9x	19.5x	70.0%	57.6%	47.6%	32.2%	39.3%	44.3%	90.6%
Asia Vital Components	3017 TT	OW	3400	41,845	32.5x	21.1x	16.7x	114.6%	54.1%	26.5%	62.1%	60.7%	46.0%	125.4%
Auras Technology	3324 TT	N	1630	4,755	28.7x	20.6x	16.5x	94.4%	39.2%	21.9%	37.7%	40.3%	36.0%	61.4%
Envicool	002837 CH	UW	56	10,588	65.9x	38.7x	27.4x	70.2%	77.2%	45.8%	25.8%	34.2%	35.7%	(32.4%)
Goaland Energy Conservation	300499 CH	NC	33	1,524	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	9.1%
Shenling Environmental Systems	301018 CH	NC	92	5,108	86.3x	47.5x	31.3x	61.4%	75.5%	57.4%	13.4%	19.4%	24.4%	101.7%
Alfa Laval	ALFA SS	N	569	23,525	26.2x	22.6x	20.7x	3.8%	15.8%	8.9%	19.5%	20.1%	19.3%	22.2%
Hisaka Works	6247 JP	NC	1648	301	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	7.2%
Dover	DOV US	NC	191	25,789	17.9x	16.4x	15.3x	9.5%	8.2%	6.2%	17.3%	17.1%	16.6%	(1.9%)
Kaori Heat Treatment	8996 TT	NC	1585	4,645	76.2x	40.0x	24.4x	132.0%	92.7%	68.1%	37.4%	47.5%	51.1%	173.3%
Tongxing Technology	301252 CH	NC	37	934	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	13.8%
Xylem	XYL US	N	101	23,561	17.8x	16.4x	14.8x	9.1%	8.9%	8.7%	12.0%	13.1%	13.6%	(25.9%)
Feilong Auto Components	002536 CH	NC	47	3,989	55.8x	35.4x	30.6x	16.8%	31.9%	38.3%	12.6%	15.4%	20.2%	56.6%
Dayuan Pumps	603757 CH	NC	69	1,929	n/a	n/a	n/a	n/a	7.2%	5.2%	8.8%	9.3%	9.7%	56.1%
Flowserve	FLS US	NC	74	9,418	18.0x	15.9x	14.7x	16.0%	12.7%	8.4%	21.9%	21.5%	19.9%	6.8%
Emerson	EMR US	OW	159	88,907	24.3x	21.9x	20.1x	8.5%	8.8%	8.0%	17.4%	17.3%	19.1%	20.1%
KITZ	6498 JP	NC	2002	1,114	12.2x	11.1x	10.9x	17.0%	9.4%	2.2%	10.8%	n/a	n/a	15.1%
Neway Valve	603699 CH	OW	44	5,069	18.2x	14.8x	12.4x	20.3%	23.5%	19.2%	34.7%	34.5%	33.5%	(16.1%)
Sanhua Intelligent Controls	002050 CH	OW	36	21,321	33.7x	28.4x	24.4x	6.7%	18.6%	16.4%	13.1%	14.1%	14.8%	(35.5%)
DunAn Environment	002011 CH	NC	13	2,003	12.3x	11.0x	10.1x	(2.4%)	11.4%	9.1%	14.6%	14.1%	13.5%	(1.4%)
Parker Hannifin	PH US	OW	970	122,272	27.5x	25.0x	22.7x	11.6%	10.1%	9.9%	26.0%	24.8%	23.5%	10.3%
Eaton	ETN US	NC	431	167,560	31.9x	27.1x	23.2x	11.1%	14.3%	14.6%	23.4%	25.1%	26.0%	35.4%
Average					29.0x	23.6x	20.4x	16.9%	18.6%	15.6%	24.4%	26.9%	26.8%	17.6%
Median					29.0x	23.6x	20.4x	16.9%	18.6%	15.6%	21.7%	22.4%	23.7%	17.6%

Note: NC=Not covered. All estimates are Bloomberg consensus.
Source: Bloomberg Finance L.P.

Industry

Genset: cooling packages for gas/diesel engines; AIDC-driven power infrastructure demand

Power infrastructure is lagging data center construction, creating a window for distributed generation

Based on the J.P. Morgan technology team's forecast, the global data center power installed base is forecast to grow from approximately 117GW in 2025 to 242GW by 2028, implying a 27% CAGR over FY25-28. Within this, the AI data center power installed base is expected to increase from around 37GW to 148GW, representing a 60% CAGR, driven by strong AI server demand and a significant increase in AI silicon power.

However, the pace of data center buildout is increasingly constrained by electricity supply. Lead times for transmission and distribution equipment in the US are rising, while concerns are growing over the grid's ability to accommodate a large amount of data center buildout. As AI data centers require reliable, around-the-clock power and delays in commissioning directly affect returns on compute infrastructure, developers are increasingly turning to behind-the-meter power solutions, using natural gas generation as either a primary power source or an interim solution before adequate grid connections become available.

Figure 25: JPM data center power market forecast (by JPM technology team)

	2022	2023	2024	2025	2026E	2027E	2028E	2025-28E CAGR
Datacenter power installed base (MW)								
Traditional server power (off premises)	74,949	76,660	79,291	80,488	82,377	84,585	93,106	5%
AI server power	2,162	6,472	17,966	36,547	63,347	105,623	148,475	60%
High end AI GPU server	1,137	4,170	13,057	25,029	42,315	69,120	92,091	54%
ASIC AI server	1,025	2,302	4,908	11,518	21,032	36,502	56,384	70%
Total Datacenter power Installed Base	77,112	83,131	97,256	117,035	145,724	190,208	241,581	27%
YoY	14%	8%	17%	20%	25%	31%	27%	

Source: IEA, J.P. Morgan estimates.

Faster delivery and modular deployment make gas engines a practical response to near-term power shortages

Gas reciprocating engines offer equipment lead times of 15-24 months, compared with 36-60 months for heavy-duty gas turbines. The shorter delivery schedules and modular design make them well suited to mid-sized campuses and phased developments, providing behind-the-meter primary or bridging power while developers await grid connections or turbine deliveries. Heavy-duty turbines, particularly in combined-cycle configurations, are better suited to long-term baseload supply at GW-scale campuses, although OEM capacity constraints are expected to persist through 2026-30. BNEF expects gas turbines to account for roughly two-thirds of onsite gas-fired generation capacity for AI data centers, leaving a meaningful opportunity for reciprocating engines.

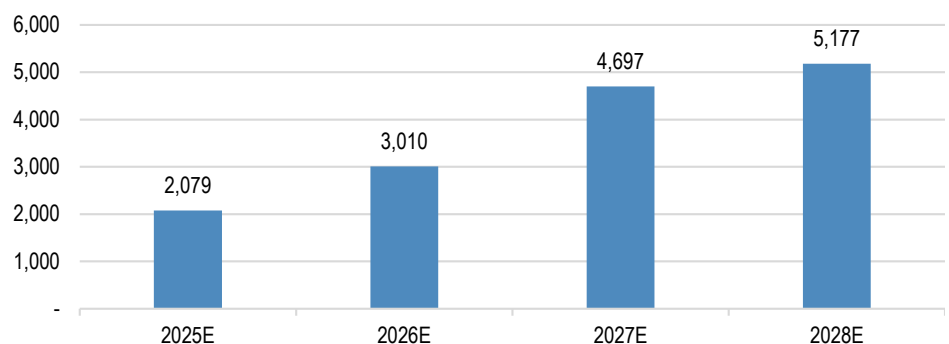
Figure 26: AI data center onsite power options: Gas reciprocating engines vs. heavy-duty gas turbines

GAS RECIPROCATING ENGINES		HEAVY-DUTY GAS TURBINES	
			
Unit size	1–20 MW	Unit size	100–1,000 MW
Equipment lead time	15–24 months	Equipment lead time	36–60 months
Plant construction	1–2 years	Plant construction	2–4 years
Start-up	<2 minutes	Start-up	5–8 minutes
Efficiency	40–50%	Efficiency	50–60% (combined cycle)
Unit cost	Rmb0.7–0.8/kWh	Unit cost	Rmb0.5–0.6/kWh
Redundancy	Lower; modular N+1	Redundancy	Higher spare capacity
Best fit	50–400 MW campuses; phased builds	Best fit	>400 MW campuses; long-term baseload
KEY ADVANTAGES	Fast deployment • Modular expansion • Lower single-point failure risk	KEY ADVANTAGES	Large scale • Higher combined-cycle efficiency • Fewer units
KEY LIMITATIONS	More units and maintenance • Lower scale efficiency	KEY LIMITATIONS	Longer delivery • Less flexible phasing • Higher outage concentration

Source: Electro Power Cell, Sina, company data, J.P. Morgan.

Driven by the growing AI data center demand for both gas reciprocating engines and backup power, we estimate the TAM for cooling packages and aftertreatment systems to expand from ~Rmb2.1bn in 2025 to Rmb5.2bn by 2028, representing a 36% CAGR.

Figure 27: AIDC genset cooling packages and after-treatment systems TAM (Rmb mn)



Source: J.P. Morgan estimates, please note 2025 numbers are based on JPM calculations.

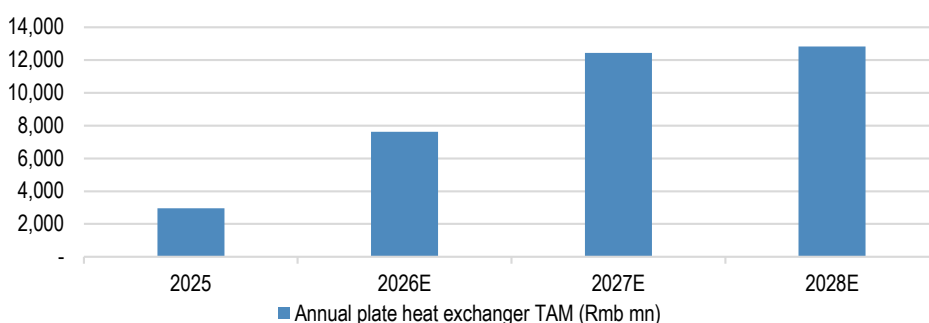
AI server liquid cooling: plate heat exchangers (PHE)

Rising AI server power density supports demand for CDUs and high-capacity heat exchangers

Higher chip power consumption and rack power density increasingly constrain air cooling in dense AI deployments, supporting greater liquid-cooling adoption and demand for coolant distribution units (CDUs). Plate heat exchangers (PHEs) are core components of liquid-to-liquid CDUs, transferring heat from the server-side coolant to the facility loop while keeping the two circuits separate.

We estimate that the global market for secondary-side plate heat exchangers (PHEs) within CDUs will grow from ~RMB3bn in 2025 to ~RMB13bn in 2028, a CAGR of approximately 63%.

Figure 28: Liquid cooling plate heat exchanger TAM estimates (JPM)



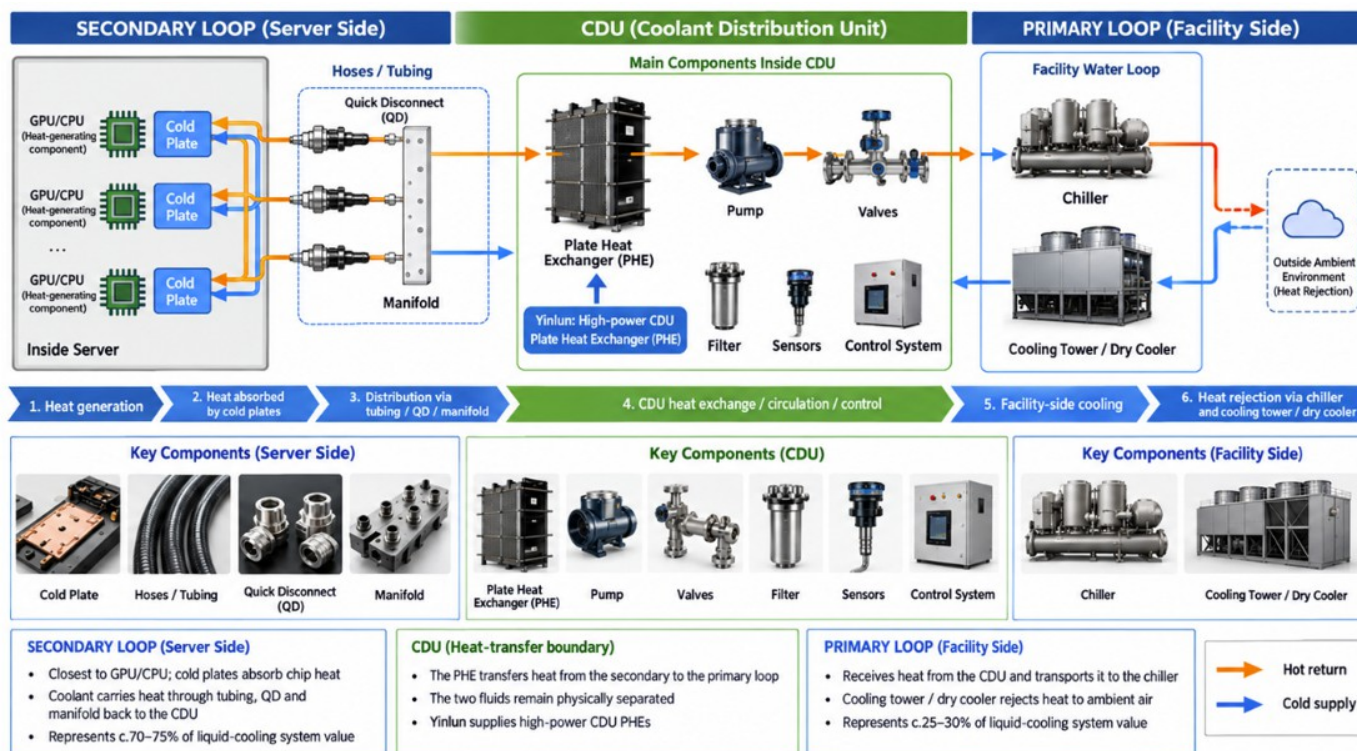
Source: Company data, J.P. Morgan estimates.

Two-Loop architecture of liquid cooling system: secondary (server-side) vs primary (facility-side)

The secondary loop, or server-side loop, is closest to the IT load. Cold plates absorb heat generated by GPUs and CPUs, while the secondary coolant carries that heat through hoses, quick disconnects (QDs) and manifolds back to the CDU. Its core role is to collect and transport heat from the servers to the CDU; it typically represents c.70-75% of total liquid cooling system value.

The primary loop, or facility-side loop, removes heat from the CDU to the outside environment. Within the CDU, the plate heat exchanger (PHE) transfers heat from the secondary coolant to the primary facility water loop. The primary loop then carries heat to the chiller and subsequently to the cooling tower or dry cooler for final heat rejection. The PHE is the boundary between the two loops: the fluids remain physically separated and exchange heat only through the metal plates. Yinlun supplies high-power CDU PHEs. The primary loop typically represents c.25-30% of system value.

Figure 29: Primary and secondary loops in a liquid cooling system



Source: J.P. Morgan, Vertiv official website

PHEs account for 10-20% of CDU costs

Plate heat exchangers represent one of the largest components of a CDU's BOM, accounting for approximately 10-20% of CDU cost, behind circulation pumps at 30-50%. Each CDU incorporates around three PHEs at an ASP of approximately Rmb40k-50k each, implying revenue content of roughly Rmb120k-150k per CDU.

Table 6: CDU BOM breakdown

Component	Function	Cost share	Technology
Pump	Circulates coolant	30–50%	Higher power, electronic control and pressure capability; motor control, high flow and sealing are key.
Plate heat exchanger	Liquid-to-liquid heat transfer	10–20%	Brazing, leak prevention and heat-transfer efficiency.
Valves	Regulates flow, pressure and temperature	10–20%	Electronic expansion and motorized regulating valves; high-end supply remains overseas-led.
Controller and inverter	Controls pumps and valves	About 10%	System-level control logic.
Sensors	Measures temperature, pressure and flow	5–10%	Accurate flow sensing supports leak detection.

Source: Company data.

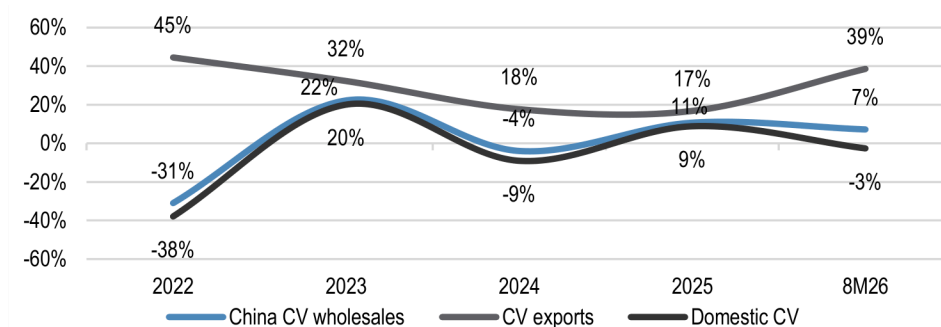
Commercial vehicles: exports and electrification trends to drive demand and content value

Export growth supports current volumes, while multiple demand drivers point to a steady, export-led up-cycle into 2027

According to CAAM data, China's commercial vehicle wholesale sales increased 7.2% y/y to 2.94mn units in 8M26, comprising domestic sales of 2.04mn units, down 2.5%,

and exports of 905k units, up 38.1%. HDTs outperformed the broader market, with sales including exports rising 16.1% to 831k units. In our view, this supports demand for Yinlun's components supplied to Chinese commercial vehicle OEMs.

Figure 30: China's commercial vehicle sales growth trend, y/y



Source: CAAM.

Looking into 2027, we expect exports to remain the primary driver of CV volume growth, with domestic demand supported by a steady HDT replacement cycle.

- **Exports:** Sustained infrastructure buildout and resource extraction activity across Africa, Southeast Asia, and Latin America continue to underpin overseas demand, while Chinese CV makers' cost-performance advantage and expanding overseas KD assembly footprints continue to take share from legacy European and American brands – supporting continued decent export growth.
- **Domestic HDT replacement demand in 2027 should be underpinned by two complementary cohorts.** China IV trucks, mostly registered in 2015-2017, will be 10-12 years old by 2027 and are already entering a high-scrappage phase; with driving restrictions and rising maintenance costs adding further pressure, this cohort's accelerated retirements driven by this year's policy support are expected to bring further replacement demand next year. China V trucks, registered mostly in 2017-2020 during the last HDT upcycle, will reach 7-10 years of age in 2027 and are also starting to enter their replacement cycle, which should contribute a degree of incremental replacement demand next year. Together, these two cohorts point to a steady replacement cycle for domestic HDT demand in 2027.

Figure 31: China EV commercial vehicle subsidy comparison, 2026E vs. 2025

Category	2025	2026	Key Changes
Purchase Tax			
Purchase tax	Full exemption (0%), no cap	50% reduction (effective 5%), no cap	Incentive reduced, but commercial vehicles remain uncapped
Trade-in Subsidy			
Eligible scrappage vehicles	China III and below diesel commercial trucks	China III and China IV diesel commercial trucks	Significantly expanded coverage, including China IV trucks
Scrappage subsidy	Rmb45,000 per vehicle	Rmb45,000 per vehicle	Unchanged
NEV HDT purchase subsidy	Rmb95,000 per vehicle	Rmb95,000 per vehicle	Unchanged
Maximum combined subsidy	Rmb140,000 per vehicle	Rmb140,000 per vehicle	Unchanged

Source: MIIT.

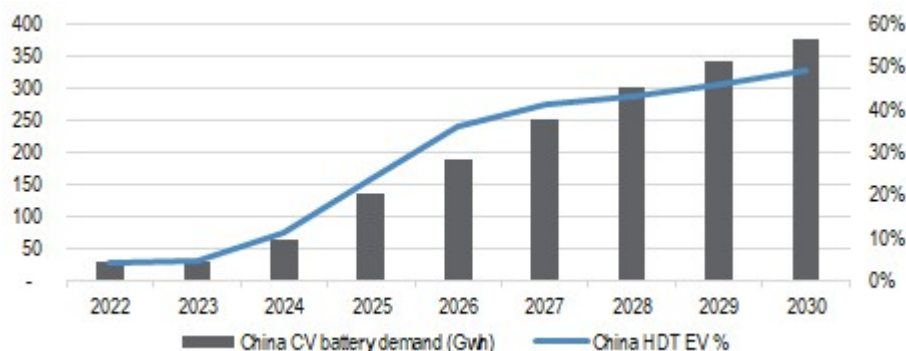
HDT electrification accelerating, backed by a new national policy roadmap

HDTs are now a meaningful swing factor within the broader CV market – accounting for ~28% of China's CV wholesales in 8M26 per CAAM data.

On 12 June 2026, China released its first national Master Plan for HDT electrification, setting a 40% new energy penetration target by 2030 – equivalent to roughly 1.6mn new energy HDTs, or ~20% of the total heavy duty truck fleet. The plan also calls for 3,000 new HDT charging/battery-swap stations, 80% electrification of fixed-route short-haul freight in key regions, and 30,000km of zero-carbon freight corridors, providing a clearer policy runway for adoption ([JPM report](#)).

We see the 40% target as a floor rather than a ceiling: China has a track record of overachieving similar policy goals (e.g., the 2020 passenger NEV plan targeted 20% penetration by 2025, which was actually surpassed by 2025), and EV HDT penetration has already tracked ahead of a linear path to the target – rising from ~5% in 2023 to ~25% in 2025 and ~33% YTD in 1H26 (with June alone reaching ~45%), aided by improving battery technology, expanding charging/swap infrastructure, and rising LNG price volatility that strengthens the TCO case for EVs. We therefore see EV HDT penetration reaching roughly 50% by 2030E, above the government's stated target.

Figure 32: China CV battery demand and HDT NEV % forecast



Source: GGII, CAAM, J.P. Morgan estimates. Please note that 2026-2030 numbers are based on JPM estimates.

Company Overview

Founded in 1958, Yinlun has more than six decades of experience in thermal-management systems. The company began as a manufacturer and repair provider for agricultural machinery and entered the commercial vehicle heat-exchanger market in 1980. Following its restructuring into a joint-stock company in 1999 and listing on the Shenzhen Stock Exchange in 2007, Yinlun expanded beyond commercial vehicles into passenger vehicles and established a global manufacturing and customer service footprint.

Yinlun has since extended its thermal-management capabilities into new energy vehicles and digital infrastructure. The company began developing NEV thermal-management systems in 2017 and subsequently expanded into data center and energy-storage liquid cooling in 2020. It established a dedicated Digital & Energy business unit in 2023, creating an operating portfolio across automotive components, genset cooling and aftertreatment systems, AI data center liquid cooling and robotics parts.

Shareholding structure

Yinlun completed a generational leadership transition in August 2026, with Xu Zhengzheng succeeding his father, Xu Xiaomin, as Chairman. Xu Xiaomin remains a director and the company's legal representative. As of end-June 2026, the two held a combined 10.35% stake, including Xu Xiaomin's 6.57% direct interest and a 3.78% stake held through Ningbo Zhengsheng Enterprise Management Partnership, and remain Yinlun's joint ultimate controllers.

Revenue breakdown and key products

Yinlun operates across three core thermal-management segments: passenger vehicles, commercial vehicles and non-road machinery, and digital and energy thermal management. In 2025, the three segments generated revenue of Rmb8.4bn, Rmb5.2bn and Rmb1.5bn, representing 53%, 33% and 9% of group revenue, respectively, with the remainder primarily from other industrial businesses. Passenger vehicles have become Yinlun's largest revenue segment, although its 2025 gross margin of 14.3% was below the 23.6% recorded by the commercial vehicle and non-road segment, reflecting more intense competition.

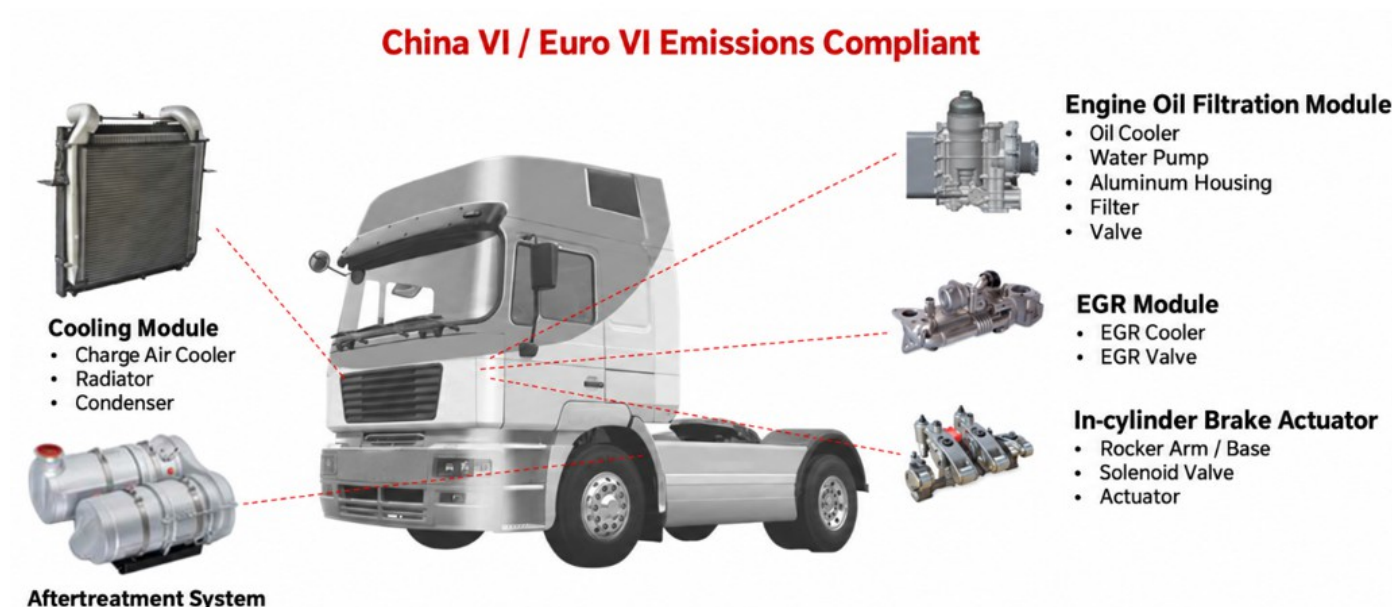
Table 7: Product portfolio, and key customers and peers

Business segment	Core products	Customers	Key peers / competitors
Passenger vehicle thermal management	Integrated thermal-management modules, HVAC, front-end cooling modules, chillers, battery cooling plates, electronic pumps/valves and fans	Ford, GM, BMW, Mercedes-Benz, Volvo, Toyota; Geely, GWM, BYD, SAIC, GAC, Li Auto, XPeng and NIO	China: Sanhua (002050 CH), Tuopu (601689 CH), Aotecar (002239 CH), Feilong (002536 CH), Yinbang (300337 CH); Global: Denso (6902 JP), Hanon Systems (018880 KS), Valeo (FR FP), Modine (MOD US)
Commercial vehicle & non-road thermal management	Cooling modules, EGR modules, oil coolers, engine brakes, HVAC systems and exhaust aftertreatment systems	Daimler Truck, Cummins, Navistar, Scania; FAW Jiefang, Sinotruk, Foton, Dongfeng, Shaanxi Auto, JAC, Weichai, Sany, XCMG and LiuGong	China: Feilong (002536 CH), Yinbang (300337 CH), Xinquan (603179 CH); Global: Modine (MOD US), T.RAD (7236 JP), Denso (6902 JP), AKG (private)
Genset cooling packages	Genset cooling packages and gas/diesel-engine aftertreatment systems	Caterpillar, Cummins; other international engine and genset OEMs under development	China: Jiangsu Weichuang (private), Weifang Xinnui (private) Global: Modine (MOD US), T.RAD (7236 JP), Dana (DAN US), MAHLE Industrial Thermal Systems (private), AKG (private)
AIRC liquid cooling	Brazed plate heat exchangers (PHEs); potential extension into CDUs, manifolds and cold plates	Luxshare, Kehua, Delta and Cooler Master as validation partners; Google represents a potential end-customer supply-chain opportunity	China: Sanhua (002050 CH), Tongxing Technology (301252 CH), Kaori (8996 TT), Jiangsu Yuanzhao (private); Global: Alfa Laval (ALFA SS), SWEP / Dover (DOV US), Danfoss (private), Kelvion (private), SPX FLOW / APV (private)
Energy-storage thermal management	Energy-storage liquid-cooling plates, liquid-cooling units and internal heat exchangers for battery energy-storage systems	CRRG, CATL and BYD	China: Envicool (002837 CH), Goaland (300499 CH), Shenling Environmental (301018 CH), Tongfei (300990 CH), Sanhua (002050 CH);

Source: Company data.

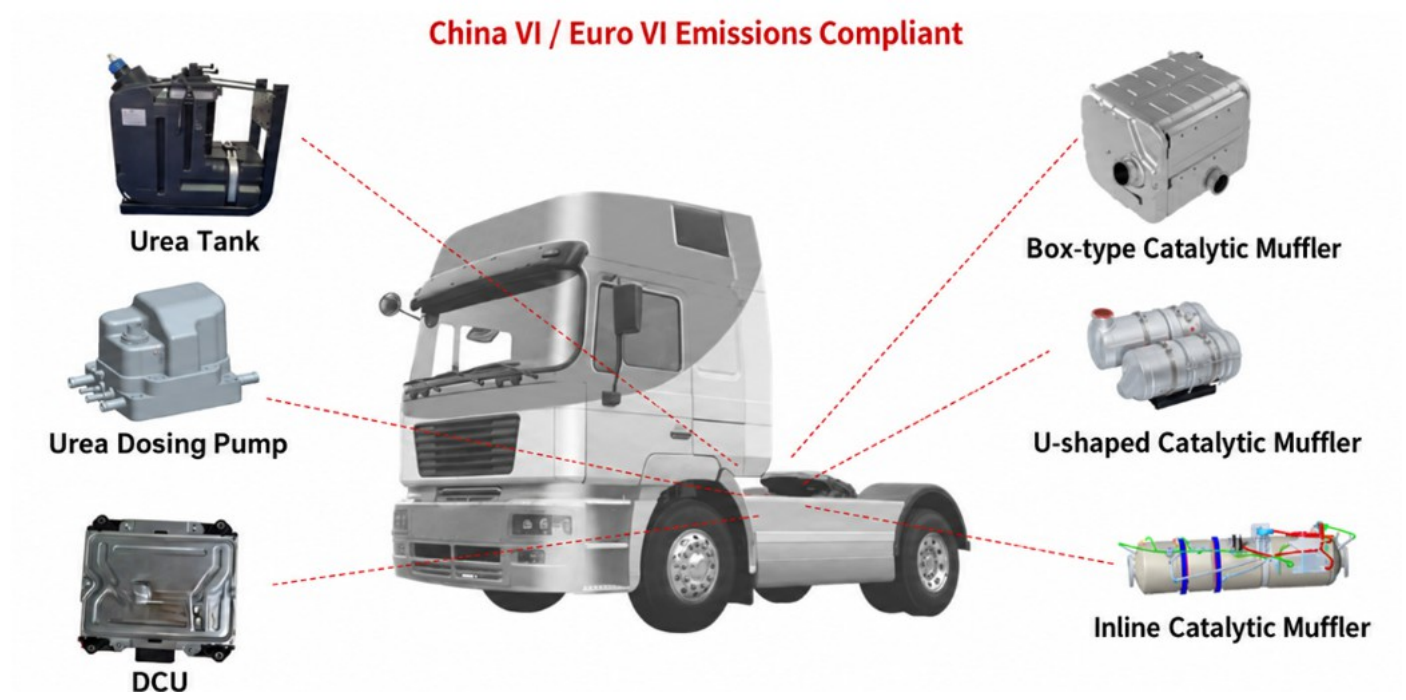
- **Commercial vehicle and non-road business:** This is Yinlun's traditional area of strength. Its key products include engine-cooling modules, radiators, charge-air coolers, oil coolers, aftertreatment systems and thermal-management systems for construction and other non-road machinery.

Figure 33: Commercial vehicle and non-road thermal management products



Source: Company data.

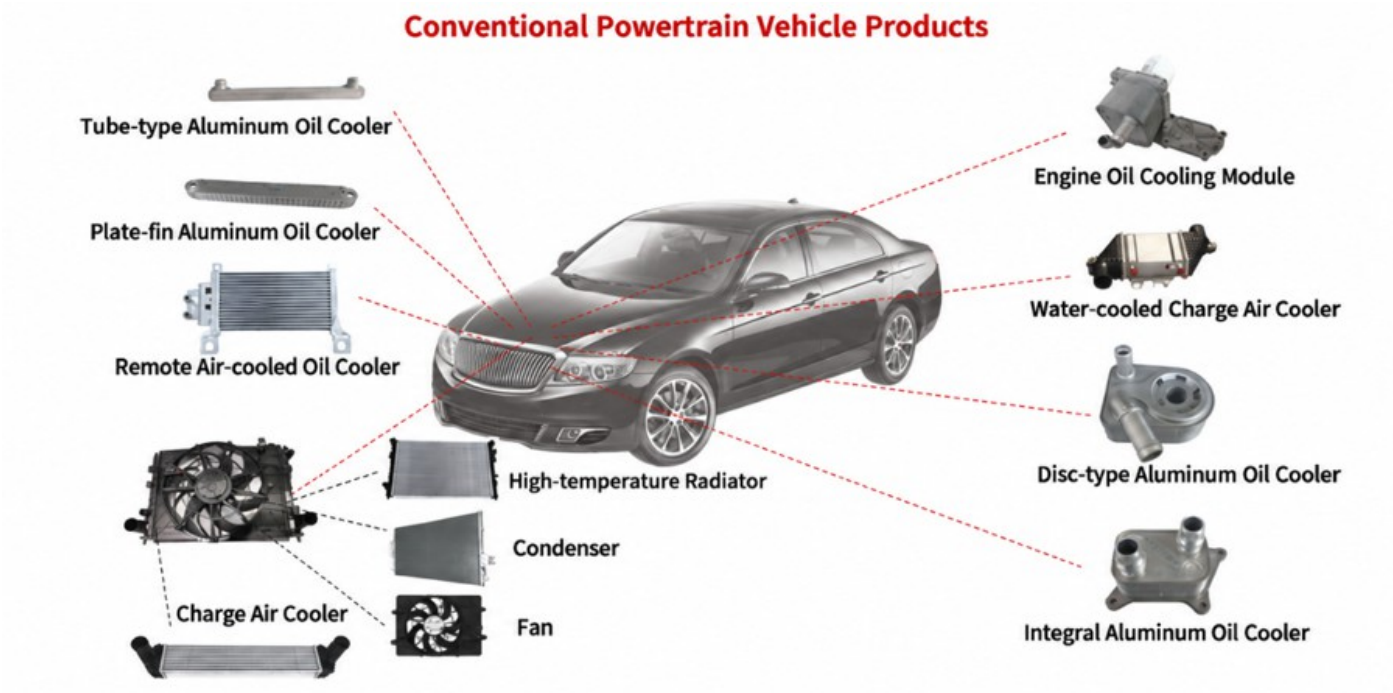
Figure 34: Commercial vehicle and non-road aftertreatment products



Source: Company data.

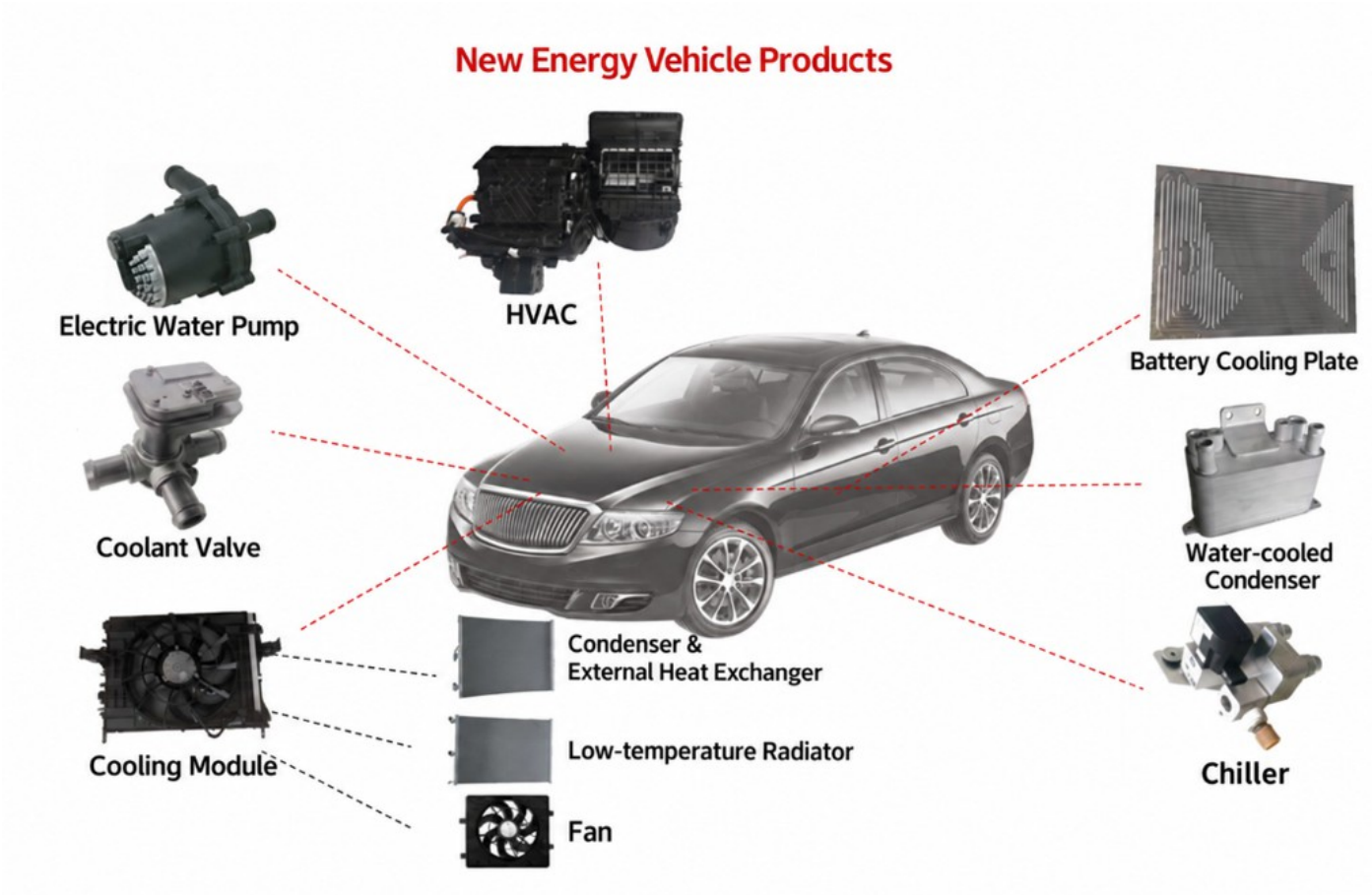
- **Passenger vehicle business:** This covers engine cooling, air conditioning and NEV thermal management. Major products include front-end cooling modules, battery cooling plates and chillers, electronic water pumps and fans, refrigerant valves and heat-pump air conditioning systems. Content per vehicle should continue to increase with vehicle electrification and greater thermal-system integration.

Figure 35: ICE passenger vehicle thermal management products



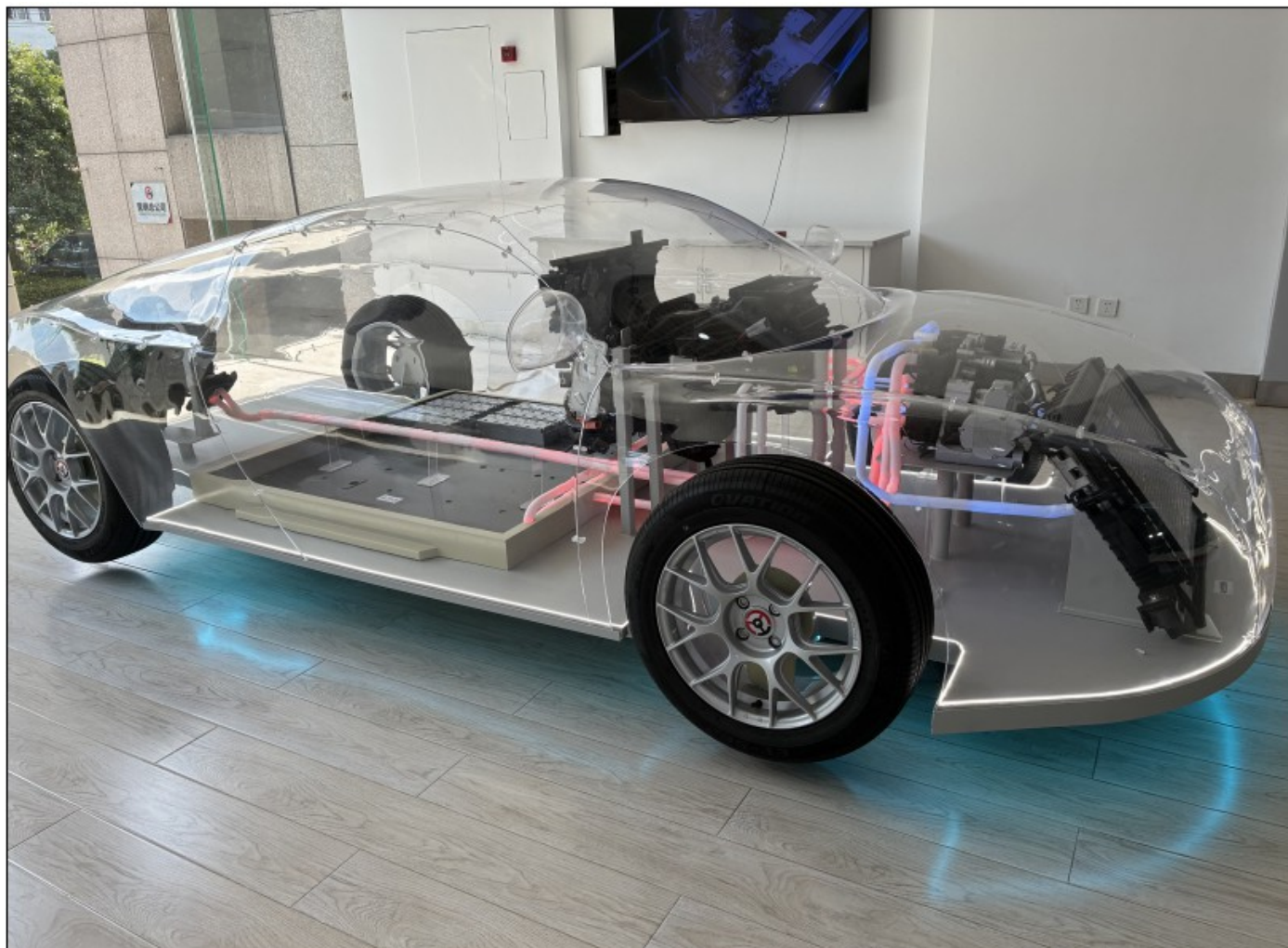
Source: Company data.

Figure 36: NEV passenger vehicle thermal management products



Source: Company data.

Figure 37: Yinlun's passenger vehicle thermal management products



Source: Company data.

Table 8: Content per car of ICE vs. NEV

Business segment	Vehicle / application	ICE/LNG content per vehicle	BEV content per vehicle
Passenger vehicle thermal management	Passenger vehicles	Rmb1,000+	Rmb3,600–3,700
Commercial vehicle thermal management	Heavy-duty trucks and construction machinery	ICE: Rmb14,000–15,000 LNG: Rmb10,000+	Rmb20,000–30,000

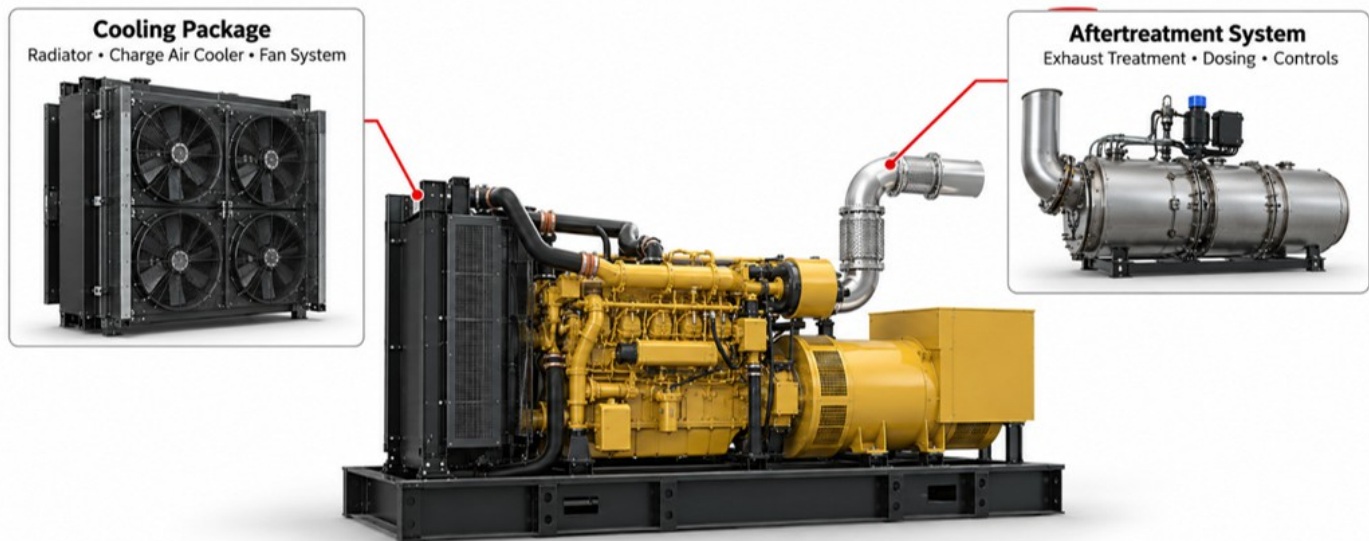
Note: ICE = Internal combustion engine; LNG = Liquid natural gas; BEV = Battery electric vehicle.

Source: Company data.

- Digital and energy business:** Extends Yinlun's capabilities in heat-exchanger design, brazing processes and large-scale manufacturing into gensets, data centers and energy storage applications. Its key products include genset cooling packages and aftertreatment systems, liquid cooling plate heat exchangers, and energy storage liquid cooling plate and heat exchangers integrated within battery energy storage systems.

- **Energy storage business (ESS):** Yinlun supplies liquid-cooling plates and thermal management units for battery energy storage systems, helping control battery temperature and support system safety, performance and cycle life. Key customers include CRRC, CATL and BYD, among others.
- **Humanoid robotics:** Yinlun is developing and supplying thermal management and structural components for humanoid robots, including motor modules, body structural parts and chest cavity cooling solutions. Its key overseas customer is Figure AI, for which management commented that Yinlun's content at c.Rmb100k per robot. Production is currently delivered from China, with US-based delivery expected to follow as overseas volumes scale. Potential domestic customer development includes Xiaomi and AgiBot (Zhiyuan). Figure AI is expected to be the largest near-term contributor, while domestic programs provide additional optionality.

Figure 38: Genset cooling package and aftertreatment system



Source: Company data.

Figure 39: Yinlun's Liquid cooling PHE



Source: J.P. Morgan.

Figure 40: Alfa Lava's liquid cooling PHE



Source: Company data.

Yinlun's 2030 targets

The company targets to achieve Rmb33bn in revenue in 2030, of which the digital and energy business would contribute Rmb10bn.

Table 9: Yinlun's revenue target in 2030 vs. 2026 JPMe

	JPMe 2026E Revenue (Rmb mn)	Revenue target (Rmb mn) (company guidance)	as % of group level revenue	2026-30E CAGR
Group level revenue	18,921	33,000	100%	15%
Digital and energy	1,938	10,000	30%	51%
Genset	538	4,000	12%	65%
Liquid cooling	500	4,000	12%	68%
ESS	900	2,000	6%	22%

Source: Company data, J.P. Morgan estimates.

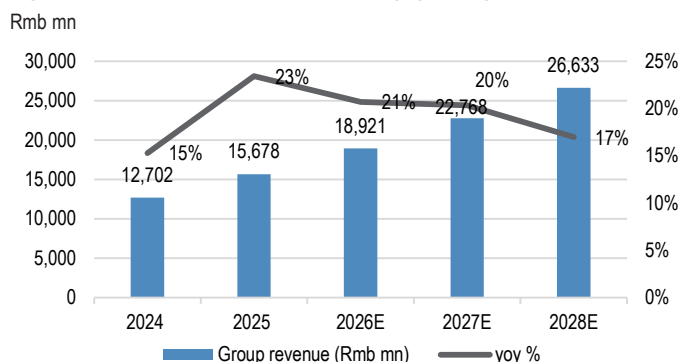
Financial Analysis

Profit and loss analysis

Revenue: three-segment structure with an accelerating mix shift toward Digital & Energy

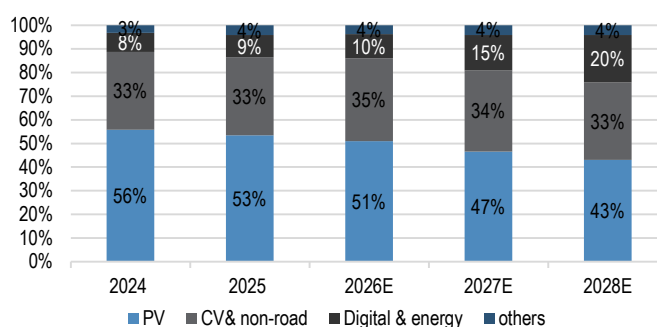
Yinlun derives revenue mainly from three segments – passenger vehicle (PV) thermal management, commercial vehicle (CV) & non-road thermal management, and digital & energy. We forecast group revenue to grow from Rmb15.7bn in 2025 to Rmb26.6bn in 2028E, at a 19% CAGR, with the digital & energy segment the key swing factor: its share of group revenue rises from 9% in 2025 to 20% in 2028E, while PV's share fades from 53% to 43% over the same period as CV & non-road holds broadly steady in the low-30s%.

Figure 41: Group revenue estimates and yoy change



Source: Company data, J.P. Morgan estimates.

Figure 42: Revenue mix by segment



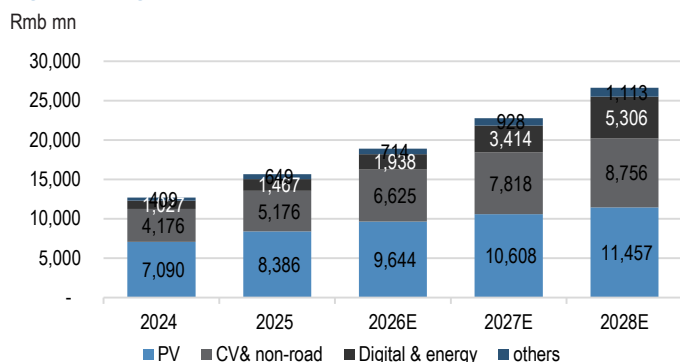
Source: Company data, J.P. Morgan estimates.

Revenue by segment

- PV growth: rapid expansion from 2021-23, becoming largest segment in 2024, with growth decelerating steadily thereafter.** Passenger vehicle thermal management revenue entered a period of rapid growth from 2021 to 2024, as NEV penetration rose and per-vehicle thermal-management content increased with electrification. PV overtook CV & non-road to become Yinlun's largest segment in 2023. From 2025, however, growth has decelerated markedly – 18% in 2025 (vs. 31% in 2024 and 50-60% in 2021-23), and we forecast a further step down to 15%/10%/8% in 2026E/27E/28E – as the segment's larger revenue base, maturing EV penetration, and intensifying pricing competition together compress both volume growth and content-per-vehicle upside.
- CV & non-road**, by contrast, reaccelerated on export demand, the domestic replacement cycle and HDT electrification since 2024, and we model growth of 28%/18%/12% in 2026E/27E/28E.
- Digital & energy** is the structural growth engine, compounding at a 54% CAGR over 2025-28E and contributing ~20% of revenue in 2028E driven by robust growth of genset-related, AIDC liquid cooling business and ESS thermal management units.
- Humanoid robotics:** We forecast robotics-related revenue of c.Rmb70-80mn in FY26E, rising to c.Rmb300mn in FY27E. Given the early stage of customer programs and uncertain production ramp schedules, we view robotics as incremental

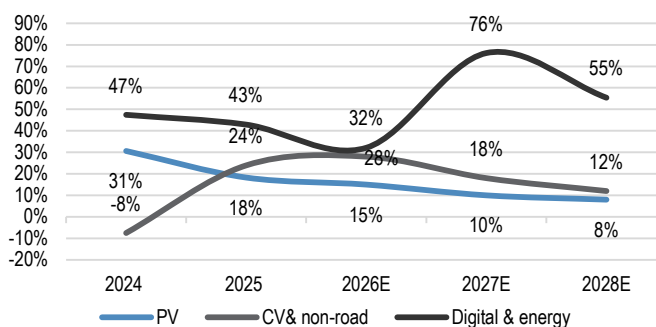
upside rather than a core earnings driver at this stage.

Figure 43: Segment revenue estimates



Source: Company data, J.P. Morgan estimates.

Figure 44: Revenue growth by segment, yoy



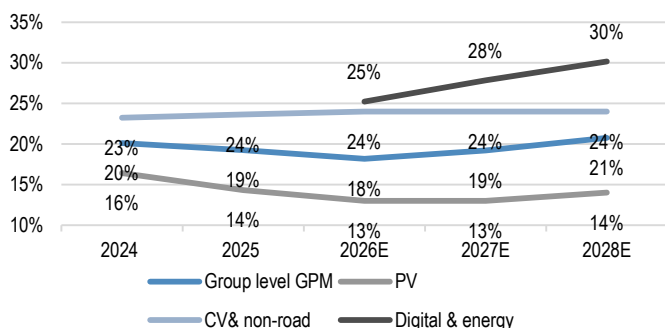
Source: Company data, J.P. Morgan estimates.

GPM

Passenger vehicle GPM fell to 13% in 1H26 vs. 14% in 2025 pressured by intensifying pricing competition and additional annual price cuts. CV & non-road GPM has held up better, staying at ~24% in 1H26 and 2025 on a more favorable export mix and electrification trend.

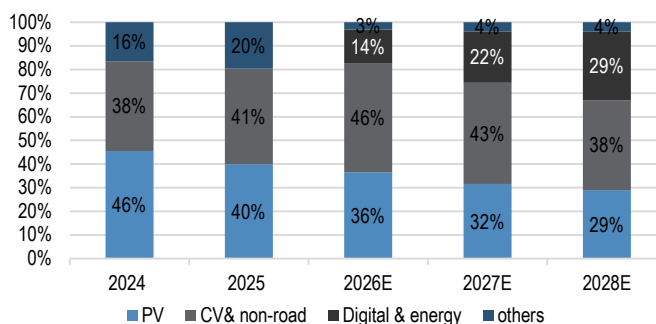
Group-level blended GPM fell to 18% in 1H26 (vs. 19% in 2025), dragged down by pressure on the PV segment's GPM and an unfavorable raw material price hike. We forecast group-level GPM to expand to ~21% by 2028E, mainly driven by the rising gross-profit contribution from the higher-margin digital & energy business.

Figure 45: GPM by segment



Note: The company did not separately disclose digital & energy segment GPM in 2024-25.
Source: Company data, J.P. Morgan estimates.

Figure 46: Gross profit mix by segment



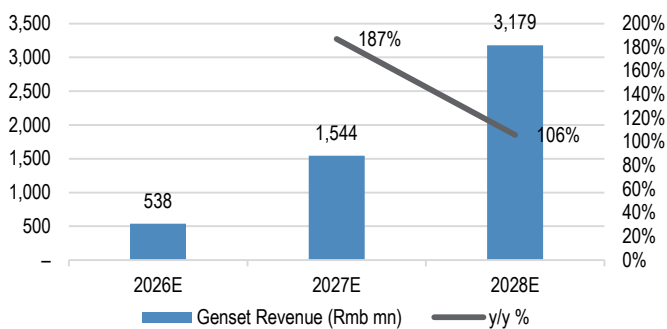
Note: The company did not separately disclose digital & energy segment GP in 2024-25.
Source: Company data, J.P. Morgan estimates.

Digital & Energy GPM

- Genset cooling packages & aftertreatment.** Yinlun began supplying Caterpillar generator-set cooling packages in 2H24, benefiting from the North American AIDC behind-the-meter power shortage. We forecast genset-related revenue to increase from Rmb538mn in 2026E to Rmb3,179mn in 2028E, a 143% CAGR, rising from 3% to 12% of group revenue. Given its structurally higher GPM at 36-38%, genset's contribution to group gross profit rises even faster, from 6% in 2026E to 21% in 2028E.

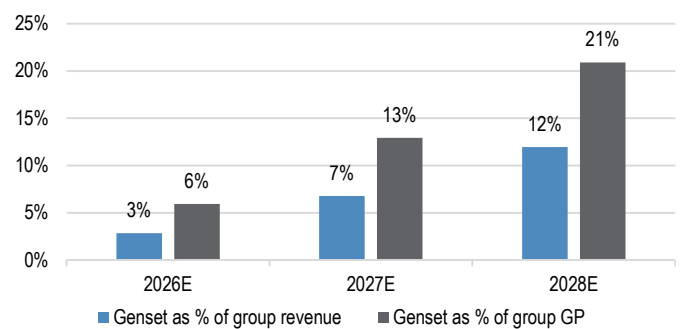
- Liquid cooling (AIDC plate heat exchangers).** In our base case (excluding potential Google orders), liquid cooling-related revenue should increase from Rmb500mn in 2026E to Rmb840mn in 2028E, at a 30% CAGR, and contributing ~3% of group revenue in 2026-28E. Given its structurally higher GPM at ~30%, liquid cooling's contribution to group gross profit rises from 4% in 2026E to 5% in 2028E. In our upside case scenario incorporating Google's PHE orders, liquid cooling revenue could reach Rmb1.4bn by 2028E (~5% of group revenue) or contribute 9% of group-level gross profit in 2028E. This could bring 0.2%/ 4%/6% net profit upside in 2026E/27E/28E, respectively.

Figure 47: Genset revenue estimates and yoy%



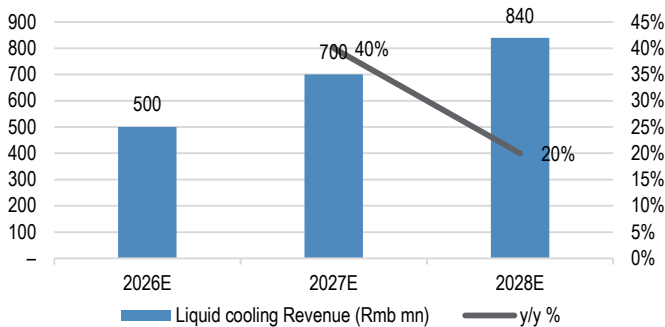
Source: J.P. Morgan estimates.

Figure 48: Genset revenue and GP contribution



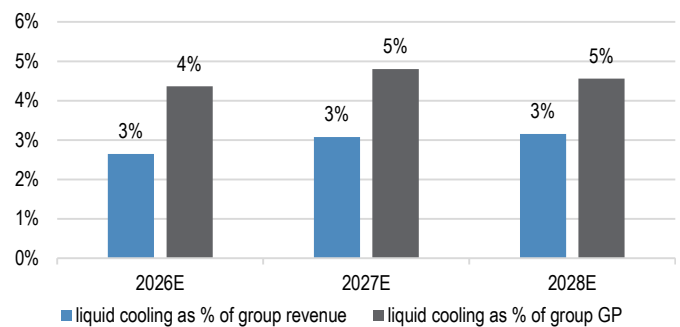
Source: J.P. Morgan estimates.

Figure 49: Liquid cooling revenue estimates and yoy% (base case)



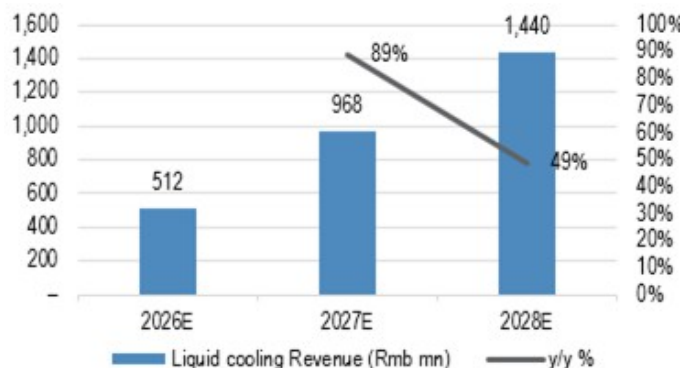
Source: J.P. Morgan estimates.

Figure 50: Liquid cooling revenue and GP contribution (base case)



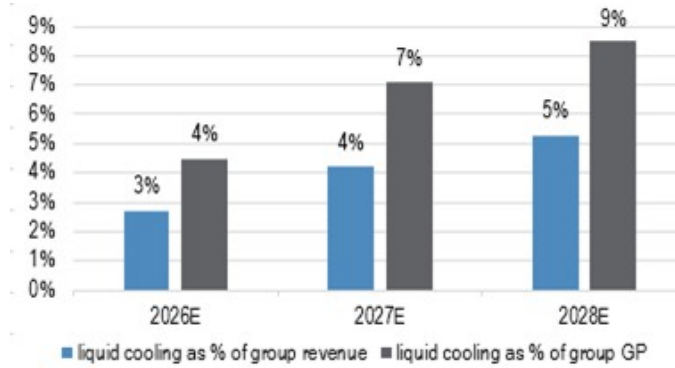
Source: J.P. Morgan estimates.

Figure 51: Liquid cooling revenue estimates and yoy% (upside scenario)



Source: J.P. Morgan estimates.

Figure 52: Liquid cooling revenue and GP contribution (upside scenario)

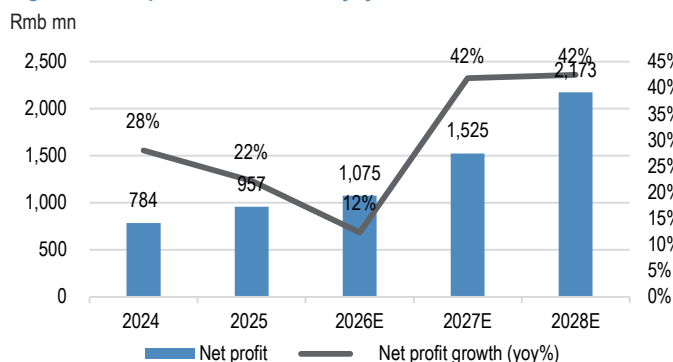


Source: J.P. Morgan estimates.

Net profit

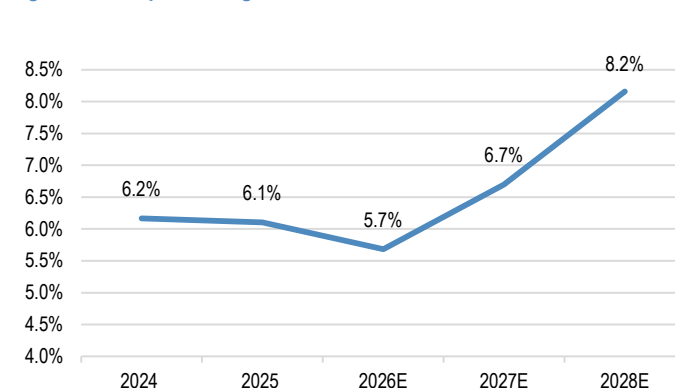
We estimate Yinlun's net profit will grow from Rmb957mn in 2025 to Rmb1,075mn in 2026E, and further to Rmb2,173mn in 2028E, at a 31% CAGR from 2025 to 2028E.

Figure 53: Net profit estimates and yoy%



Source: J.P. Morgan estimates.

Figure 54: Net profit margin estimates



Source: J.P. Morgan estimates.

Balance sheet and cash flow analysis

Balance sheet analysis

- DuPont: ROE improvement is increasingly earnings-driven.** We estimate Yinlun's ROE rises from 12% in 2023 to 21% by 2028E, with the composition shifting favorably – higher net margin (from 6% in 2023 to 8% in 2028E) and faster asset turnover (from 0.7x in 2023 to 0.9x in 2028E) thanks to higher-margin digital & energy business scale.

Table 10: DuPont analysis

	2023	2024	2025	2026E	2027E	2028E
Net margin	6%	6%	6%	6%	7%	8%
Asset turnover (x)	0.7	0.7	0.8	0.8	0.9	0.9
Equity multiplier (x)	2.9	3.0	3.0	2.9	2.8	2.7
ROE	12%	13%	14%	14%	17%	21%

Source: Company data, J.P. Morgan estimates.

- **Financial structure:** leverage peaked around 2023-25 and is set to decline steadily as capex intensity moderates relative to the growing earnings base. We estimate total liabilities/total assets falls from 62% in 2025 to 58% by 2028E, and net debt/equity nearly halves from 55% (2025) to 20% (2028E), as the favorable input-output ratio and scaling cash generation of the digital & energy business increasingly self-funds capacity expansion, reducing reliance on incremental borrowing.

Figure 55: Financial structure analysis

	2023	2024	2025	2026E	2027E	2028E
total liability / total asset	62%	61%	62%	61%	59%	58%
total liability / total equity	165%	160%	163%	155%	145%	137%
LT debt / equity	5%	2%	4%	3%	3%	2%
Net Debt/Equity	66%	56%	55%	51%	37%	20%
int. bearing debt / equity	97%	87%	86%	82%	74%	67%

Source: Company data, J.P. Morgan estimates.

Cash flow analysis

We estimate Yinlun's operating cash flow will rise from Rmb1,405mn in 2025 to Rmb3,405mn in 2028E, supported by solid earnings growth as the higher-margin digital & energy business scales. Given the favorable input-output ratio of the digital & energy business, capex stays modest relative to operating cash flow (Rmb1,180mn-1,080mn p.a. in 2026-27E), allowing FCF to rise to Rmb2,425mn by 2028E as capex needs ease even as operating cash flow continues to grow.

Table 11: Cash flow analysis

Rmb mn	2023	2024	2025	2026E	2027E	2028E
OP cash flow	921	1,205	1,405	1,649	2,492	3,405
Capex	(1,013)	(824)	(1,197)	(1,180)	(1,080)	(980)
FCF	(92)	381	208	468	1,411	2,425

Source: Company data, J.P. Morgan estimates. FCF= OP cash flow - capex

Investment Thesis, Valuation and Risks

Yinlun - A (Overweight; Price Target: Rmb51.00)

Investment Thesis

Yinlun is a China-based thermal-management supplier with more than four decades of experience in heat exchanger design, brazing and large-scale manufacturing. Its established passenger vehicle and commercial vehicle businesses provide a broad revenue base, while its global manufacturing footprint and long-standing relationships with domestic and international OEMs support expansion into adjacent industrial applications. The key upside comes from AI infrastructure. Yinlun is extending its thermal-management capabilities into data center power generation and liquid cooling, two markets with structurally faster growth and stronger margin potential than its traditional auto parts business. Caterpillar genset cooling packages provide the clearest near-term earnings catalyst, supported by rising AI- and data center-driven power demand and a multi-year supplier qualification barrier. Liquid-cooling PHEs represent a second growth opportunity as higher AI server power density drives CDU demand; Google-related qualification remains potential upside rather than a base-case assumption. Together, these businesses should support a material improvement in Yinlun's earnings mix and valuation profile.

Valuation

We have a Dec-27 price target of Rmb51. We use an SOTP valuation, applying a 15x 2028E P/E to the PV and CV auto parts segments and a 35x 2028E P/E to the Digital & Energy segment, reflecting premium growth-stage multiples for AIDC power infrastructure and liquid cooling exposure. This implies a blended 29x 2027E P/E and 20x 2028E P/E.

SOTP valuation

Segment	2028E EPS (Rmb)	P/E multiple	Value p/sh (Rmb)
PV and CV auto parts	1.9	15	28.2
Digital and energy	0.6	35	22.5
Total PT			51
Implied 2028E P/E			20

Source: J.P. Morgan estimates.

Risks to Rating and Price Target

The key risks are customer concentration in the genset business and slower-than-expected liquid-cooling order conversion. Near-term genset growth is highly dependent on Caterpillar, leaving Yinlun exposed to changes in customer procurement, supply share or pricing. In liquid cooling, PHE projects remain at the qualification and validation stage; delays in Tier-1 certification or order conversion could defer revenue and reduce utilization of newly added capacity. And geopolitical risk - while Yinlun's thermal-management components are not currently direct targets of US export controls or power-infrastructure reviews, a broader shift toward localized sourcing could raise compliance requirements, lengthen customer qualification cycles and limit new overseas order opportunities.

Yinlun - A: Summary of Financials

Income Statement						Cash Flow Statement					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	12,702	15,678	18,921	22,768	26,633	Cash flow from operating activities	1,205	1,405	1,649	2,492	3,405
COGS	(10,147)	(12,662)	(15,481)	(18,395)	(21,104)	o/w Depreciation & amortization	642	752	761	806	831
Gross profit	2,555	3,015	3,440	4,374	5,529	o/w Changes in working capital	(582)	(743)	(373)	(101)	48
SG&A	(860)	(1,008)	(1,192)	(1,434)	(1,678)	Cash flow from investing activities	(906)	(1,325)	(1,180)	(1,080)	(980)
Adj. EBITDA	1,651	2,020	2,196	2,836	3,718	o/w Capital expenditure	(824)	(1,197)	(1,180)	(1,080)	(980)
D&A	(642)	(752)	(761)	(806)	(831)	as % of sales	6.5%	7.6%	6.2%	4.7%	3.7%
Adj. EBIT	1,009	1,269	1,434	2,030	2,887	Cash flow from financing activities	(130)	309	(112)	(159)	(435)
Net Interest	-	-	-	-	-	o/w Dividends paid	(200)	(156)	(112)	(159)	(435)
Adj. PBT	1,002	1,253	1,416	2,008	2,861	o/w Shares issued/(repurchased)	345	117	0	0	0
Tax	(108)	(167)	(189)	(268)	(382)	o/w Net debt issued/(repaid)	(151)	469	0	0	0
Minority Interest	(111)	(129)	(152)	(215)	(306)	Net change in cash	176	388	356	1,252	1,990
Adj. Net Income	784	957	1,075	1,525	2,173	Adj. Free cash flow to firm	131	(102)	435	1,364	2,377
Reported EPS	0.92	1.11	1.25	1.77	2.52	y/y Growth	(132.6%)	(177.9%)	(527.4%)	213.5%	74.2%
Adj. EPS	0.92	1.11	1.25	1.77	2.52						
DPS	0.12	0.12	0.13	0.19	0.52						
Payout ratio	13.0%	10.8%	10.8%	10.8%	20.6%						
Shares outstanding	852	862	862	862	862						
Balance Sheet						Ratio Analysis					
	FY24A	FY25A	FY26E	FY27E	FY28E		FY24A	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	2,187	2,555	2,911	4,163	6,153	Gross margin	20.1%	19.2%	18.2%	19.2%	20.8%
Accounts receivable	4,637	5,657	6,824	7,533	8,632	EBITDA margin	13.0%	12.9%	11.6%	12.5%	14.0%
Inventories	2,273	2,786	3,093	3,679	4,221	EBIT margin	7.9%	8.1%	7.6%	8.9%	10.8%
Other current assets	2,396	3,104	3,016	3,137	3,277	Net profit margin	6.2%	6.1%	5.7%	6.7%	8.2%
Current assets	11,494	14,102	15,845	18,512	22,283	ROE	13.4%	14.1%	13.7%	16.9%	20.6%
PP&E	3,857	4,761	5,139	5,373	5,481	ROA	4.5%	4.8%	4.7%	6.0%	7.5%
LT investments	365	237	237	237	237	ROCE	7.6%	8.2%	8.1%	10.4%	13.1%
Other non current assets	2,646	2,777	2,818	2,859	2,900	SG&A/Sales	6.8%	6.4%	6.3%	6.3%	6.3%
Total assets	18,362	21,876	24,038	26,980	30,900	Net debt/Equity	0.6	0.6	0.5	0.4	0.2
Short term borrowings	2,064	2,455	2,455	2,455	2,455	Net debt/EBITDA	2.4	2.3	2.2	1.4	0.7
Payables	4,035	5,309	5,712	6,502	7,690	Sales/Assets (x)	0.7	0.8	0.8	0.9	0.9
Other short term liabilities	3,811	4,369	5,013	5,584	6,272	Assets/Equity (x)	3.0	3.0	2.9	2.8	2.7
Current liabilities	9,909	12,132	13,179	14,540	16,416	Interest cover (x)	-	-	-	-	-
Long-term debt	129	317	317	317	317	Operating leverage	154.7%	110.0%	63.1%	204.4%	248.7%
Other long term liabilities	1,252	1,124	1,124	1,124	1,124	Tax rate	10.8%	13.4%	13.4%	13.4%	13.4%
Total liabilities	11,290	13,572	14,620	15,980	17,856	Revenue y/y Growth	15.3%	23.4%	20.7%	20.3%	17.0%
Shareholders' equity	6,213	7,357	8,320	9,686	11,424	EBITDA y/y Growth	22.0%	22.4%	8.7%	29.2%	31.1%
Minority interests	859	947	1,099	1,314	1,620	EPS y/y Growth	26.0%	20.7%	12.3%	41.8%	42.5%
Total liabilities & equity	18,362	21,876	24,038	26,980	30,900						
BVPS	7.30	8.53	9.65	11.23	13.25	Valuation					
y/y Growth	12.1%	16.9%	13.1%	16.4%	17.9%		FY24A	FY25A	FY26E	FY27E	FY28E
Net debt/(cash)	3,958	4,581	4,798	4,025	2,630	P/E (x)	37.2	30.8	27.4	19.3	13.6
						P/BV (x)	4.7	4.0	3.5	3.0	2.6
						EV/EBITDA (x)	21.8	18.1	16.9	12.9	9.5
						Dividend Yield	0.4%	0.4%	0.4%	0.6%	1.5%

Source: Company reports and J.P. Morgan estimates.

Note: Rmb in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Yinlun - A or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Yinlun - A or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Yinlun - A or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Yinlun - A or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Yinlun - A or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Yinlun - A (002126.SZ, 002126 CH) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's

team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Wen, Rebecca Y: CALB (3931.HK), CATL - A (300750.SZ), CATL - H (3750.HK), Dynanonic - A (300769.SZ), EVE Energy (300014.SZ), Fuyao Glass - A (600660.SS), Fuyao Glass - H (3606.HK), Gotion High-Tech (002074.SZ), Hunan Yuneng - A (301358.SZ), Minth (0425.HK), Nexteer (1316.HK), Ningbo Joyson (600699.SS), Ningbo Tuopu (601689.SS), Putailai (603659.SS), Ronbay - A (688005.SS), Wuxi Lead - A (300450.SZ), Wuxi Lead - H (470.HK), Yunnan Energy (002812.SZ)

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to

be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Digital assets, which may also be referred to as virtual or crypto assets or crypto currencies (e.g. bitcoin), discussed in this research report are subject to a rapidly changing regulatory landscape. The application of relevant laws in respect of such assets may be unsettled, and the legality of any digital asset investment may be unclear. Prior to making any investment decision regarding any digital asset, investors should seek their own legal advice and be aware of any legal considerations regarding the purchase, holding or selling of such an investment. For relevant regulatory advisories on digital assets, including bitcoin and ethereum, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpil.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpil.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpil.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 070/08/2026 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results.

Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Any mentions of third-party trademarks, brand names, products, and services are for referential purposes only and any mention thereof is not meant to imply any sponsorship, endorsement, or affiliation.

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised August 29, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 29 Sep 2026 11:30 PM HKT

Disseminated 29 Sep 2026 11:31 PM HKT